

Supplementary materials

Plank et al.

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S1 Package versions

The following packages are used in this RMarkdown file:

```
## [1] "R version 4.5.2 (2025-10-31)"
## [1] "knitr version 1.51"
## [1] "ggplot2 version 4.0.2"
## [1] "brms version 2.23.0"
## [1] "designr version 0.1.13"
## [1] "bridgesampling version 1.2.1"
## [1] "tidyverse version 2.0.0"
## [1] "ggpubr version 0.6.2"
## [1] "ggpattern version 1.2.1"
## [1] "gggrain version 0.1.0"
## [1] "bayesplot version 1.15.0"
## [1] "SBC version 0.3.0.9000"
## [1] "rstatix version 0.7.3"
## [1] "BayesFactor version 0.9.12.4.7"
## [1] "effectsize version 1.0.1"
## [1] "bayestestR version 0.17.0"
```

S2 Methods

S2.1 Preprocessing of pupil sizes

We classified any extreme values (above 8 and below 2mm) as well as participant-specific extreme velocities and blinks as artefacts. Then, we low-pass filtered the data with a threshold of 4Hz based on Kret and Sjak-Shie (2019), smoothed the data and interpolated all artefacts shorter than half a second with cubic spline interpolation. Next, data was downsampled to 100Hz and trials with more than one third of the datapoints missing were excluded from analysis. Last, we applied correction based on a baseline of 200ms before onset of the face presentation. We visualised data quality and excluded trials if the pupil trace indicates recording artefacts or baseline artefacts as well as if the baseline pupil size is considered an outlier (absolute z-score larger than 2).

S2.2 Bayesian linear (mixed) models

All contrasts are set to sum coding. The models for the volatilities extracted from the HGF and for the pupil sizes used a Gaussian likelihood, while both the reaction times and learning rate updates were modelled using a lognormal distribution. The reaction time model additionally included a shift. All models included the predictor Group (ASD, COMP). The reaction time model also included Expectancy, Phase and Difficulty as well as all interactions on the population-level, and participant with slopes for Expectancy, Phase, Difficulty and their interactions as well as trials with slopes for Group on the group-level. The hypothesis regarding pupil sizes was evaluated using a model with the predictors Group, Expectancy and their interaction as well as intercepts for the participants on the group-level. Since pupil sizes are sensitive to physical movements (Mathôt & Vilotijević, 2023), we include RT of each trial instead of including trial on the group-level, as preregistered. The model for the learning rate updates included Group, Level (environmental or cue-outcome association) and Change (stable to volatile or volatile to stable) on the population-level as well as participants with slopes for Level and Change on the group-level.

S3 Model evaluation

S3.1 Parameter recovery and model identifiability

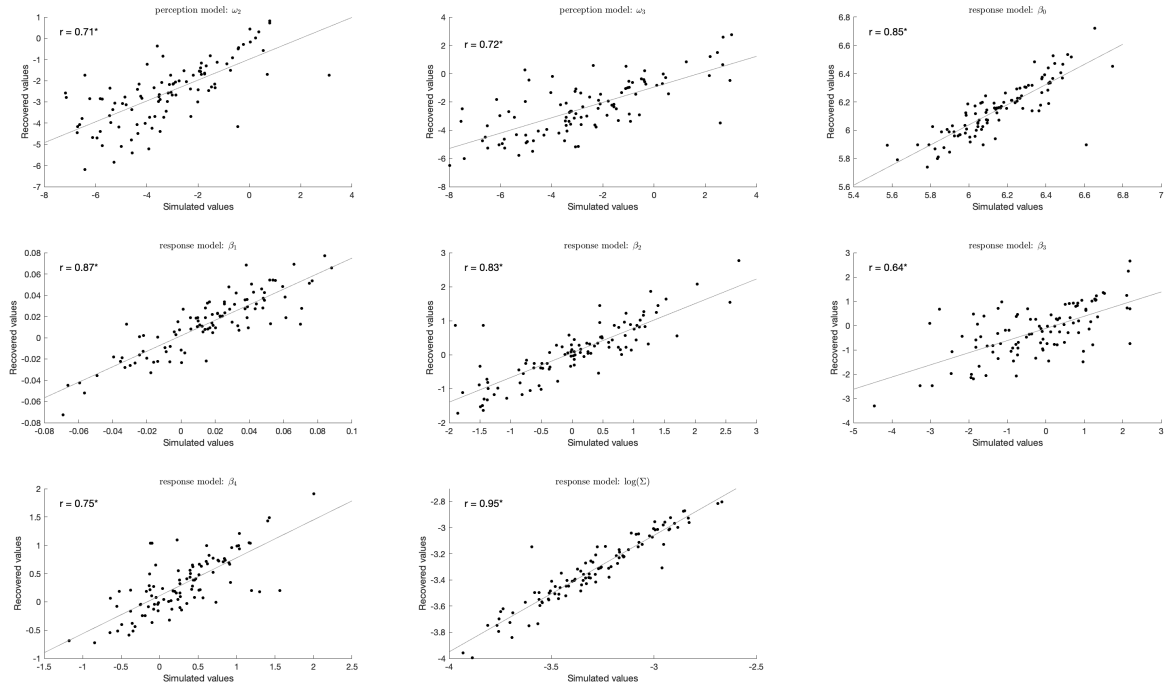


Figure 1: Enhanced HGF with empirical priors

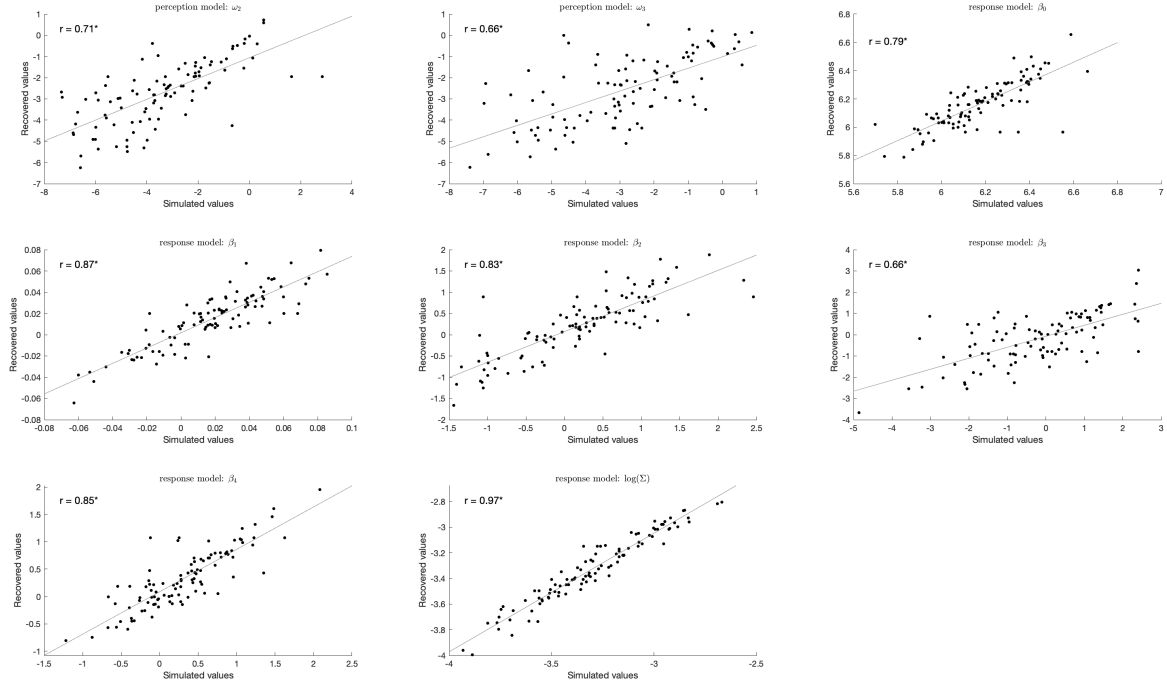


Figure 2: Original HGF with empirical priors

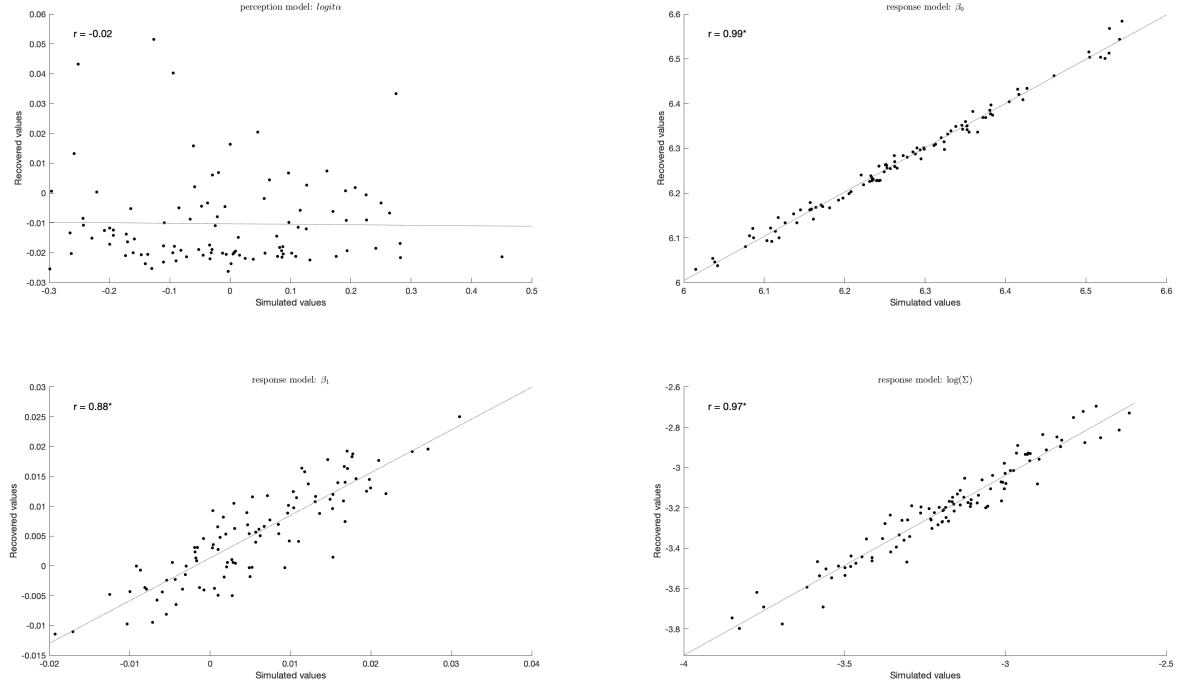


Figure 3: Rescorla-Wagner with empirical priors

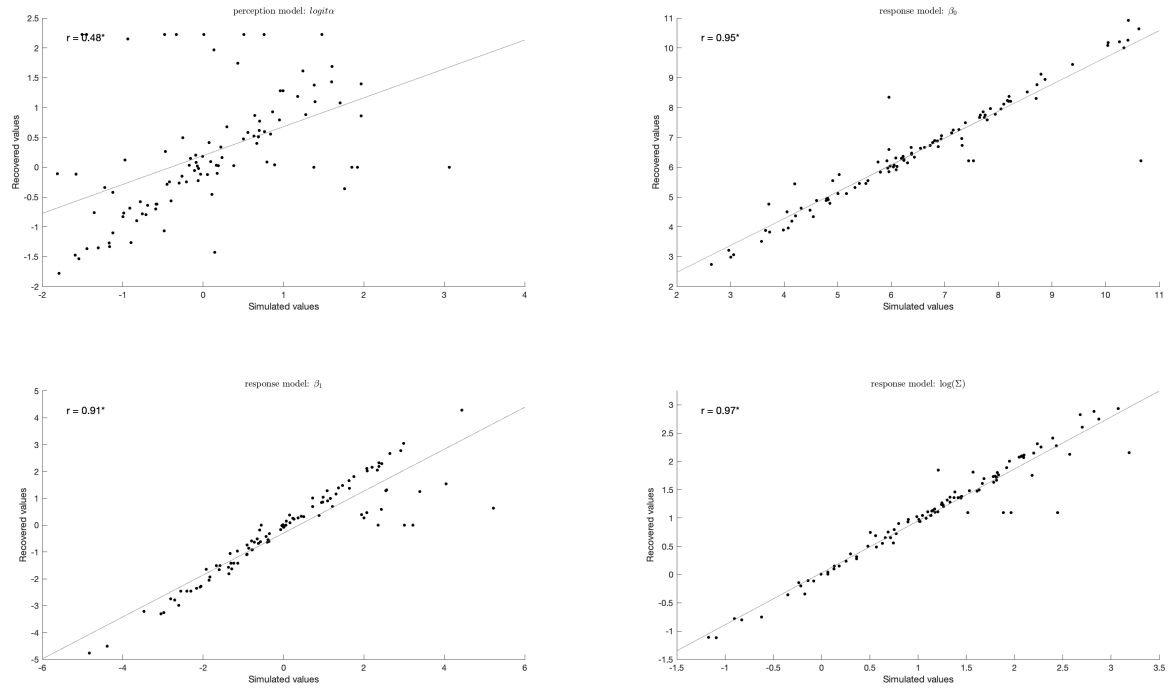


Figure 4: Model identifiability

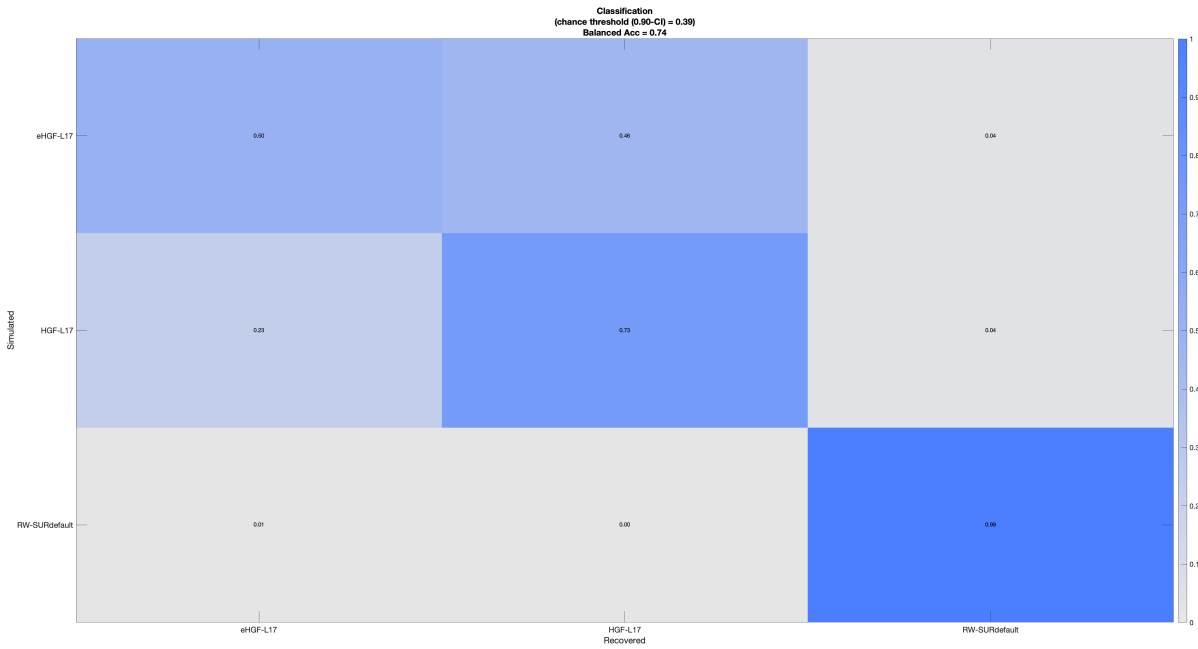
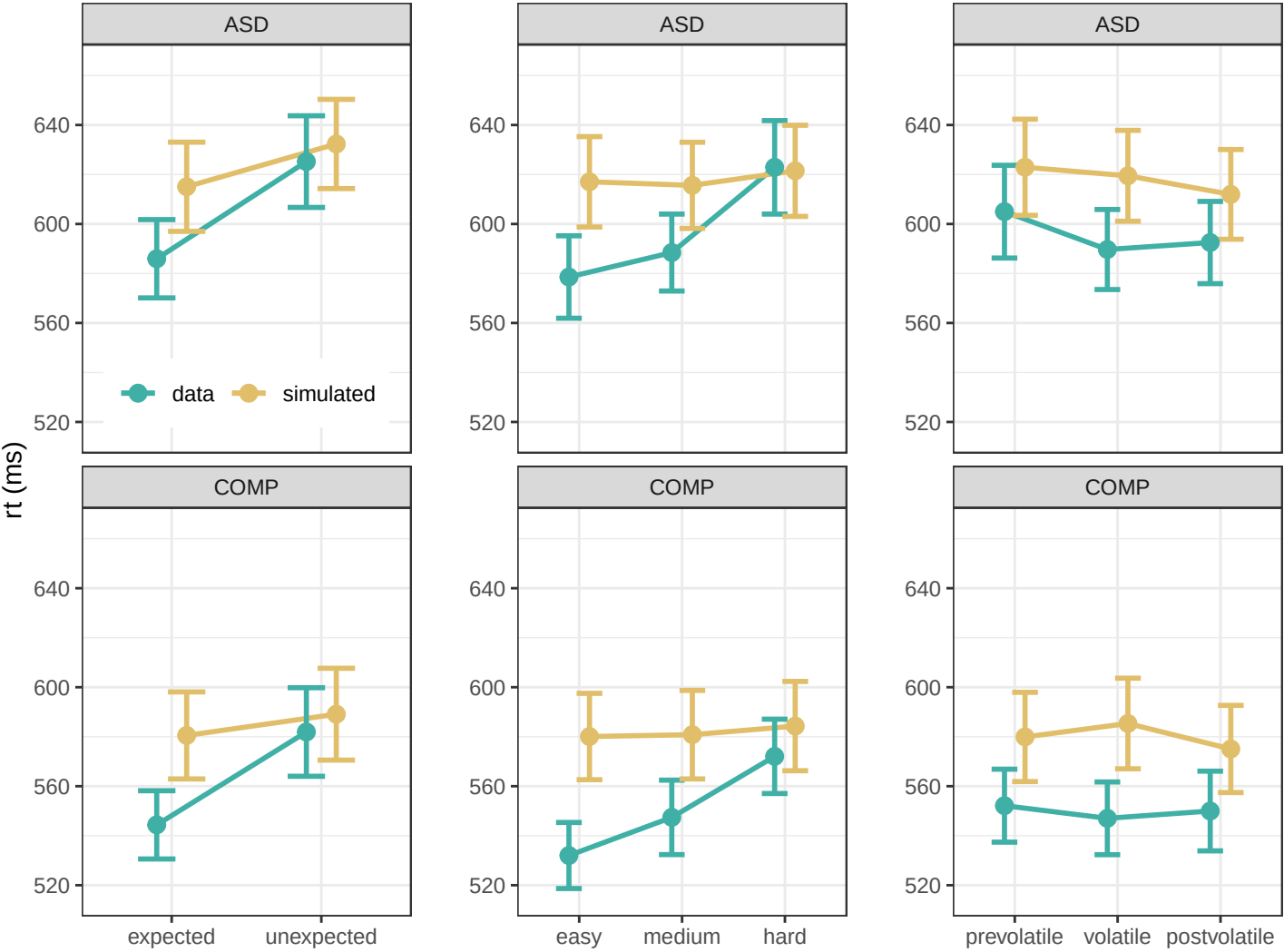
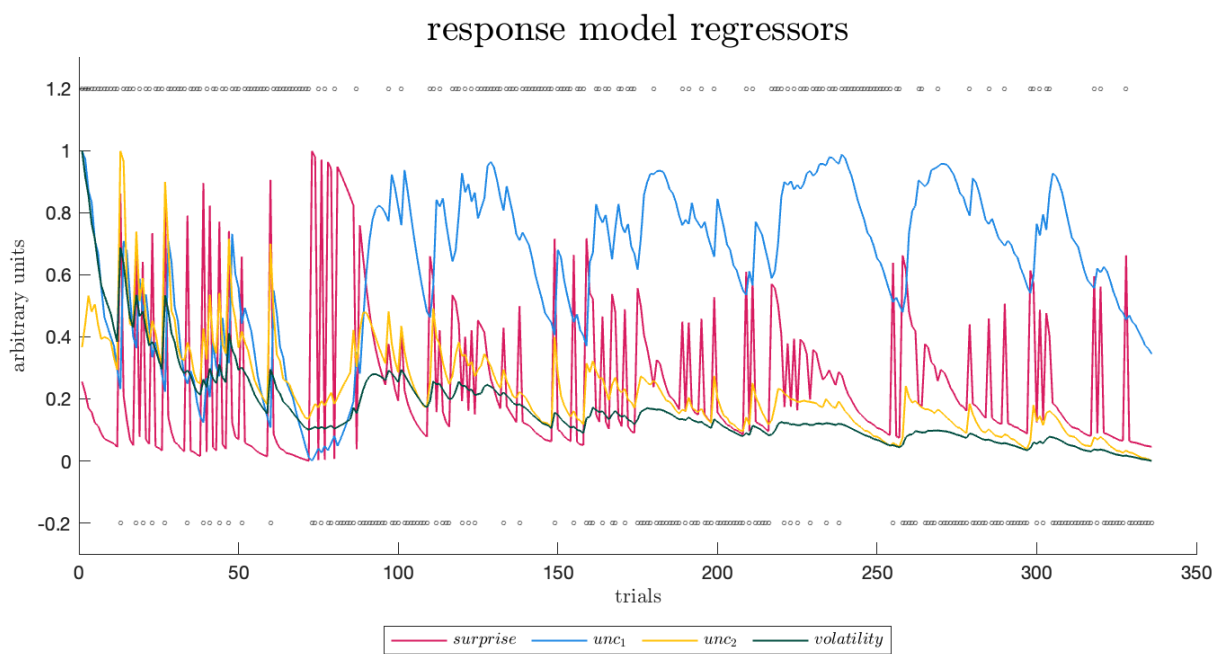
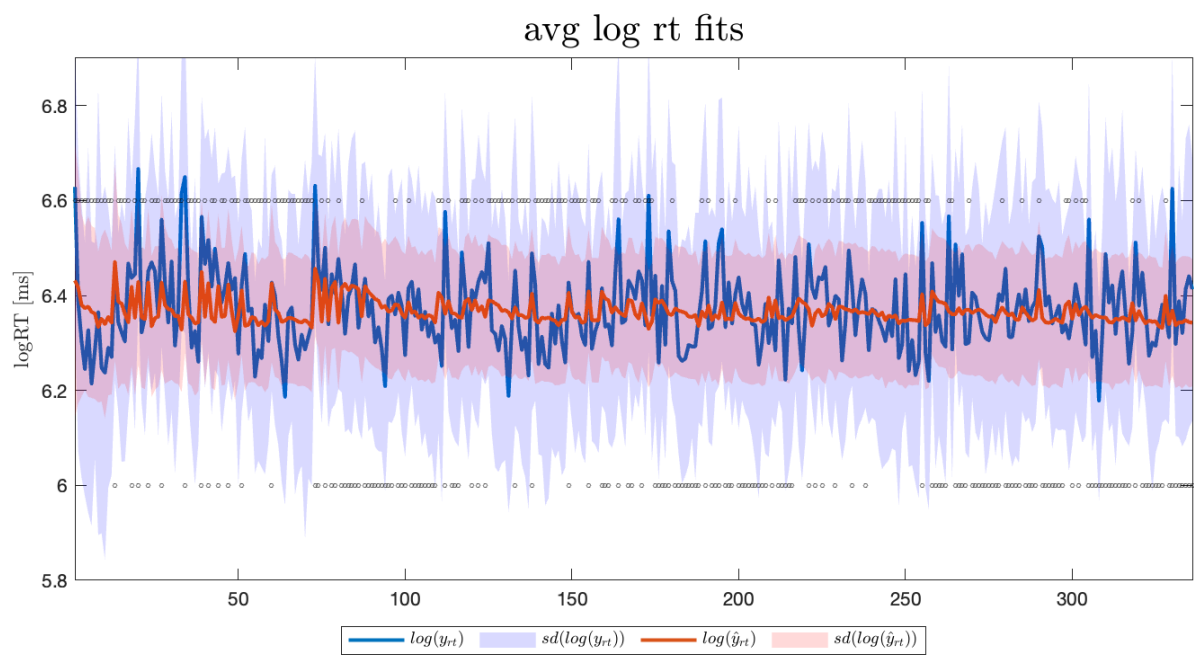


Figure 5: Rescorla-Wagner with empirical priors

S3.2 Posterior predictive checks for oHGF

Posterior predictive checks: HGF





S3.3 Comparison of LME across groups

```
## ---
## Model:
## Type: BFlinearModel, JZS
## Intercept only
## Data types:
## diagnosis : fixed
```

	bf	error	time	code
diagnosis	-1.04496	5.81e-05	Wed Oct 15 12:08:23 2025	144d96dd874c4

S4 Volatility parameters and learning rate updates

S4.1 Phasic volatility

```
## Family: gaussian
## Links: mu = identity
## Formula: be4 ~ diagnosis
## Data: df.hgf (Number of observations: 44)
## Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
## total post-warmup draws = 8000
##
## Regression Coefficients:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept      0.15      0.07   0.01   0.30 1.00    6857    4868
## diagnosis1     0.10      0.07  -0.03   0.24 1.00    7164    5392
##
## Further Distributional Parameters:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma      0.48      0.05   0.39   0.60 1.00    6978    5752
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

S4.2 Environmental tonic volatility

```
## Family: gaussian
## Links: mu = identity
## Formula: om3 ~ diagnosis
## Data: df.hgf (Number of observations: 44)
## Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
## total post-warmup draws = 8000
##
## Regression Coefficients:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept    -2.25      0.22  -2.68  -1.81 1.00    7060    5591
## diagnosis1   -0.03      0.16  -0.36   0.29 1.00    7761    5723
##
## Further Distributional Parameters:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma      1.46      0.13   1.22   1.74 1.00    7903    5912
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

S4.3 Bernoulli model with HGF parameters

```
## Family: bernoulli
## Links: mu = logit
## Formula: group ~ sbe1 + sbe2 + sbe3 + sbe4 + sze + som2 + som3
## Data: df.hgf (Number of observations: 44)
## Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
## total post-warmup draws = 8000
##
## Regression Coefficients:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept      0.00      0.27   -0.52    0.53 1.00   11653    6220
## sbe1          -0.46      0.42   -1.30    0.35 1.00    7619    6607
## sbe2           0.28      0.40   -0.48    1.09 1.00    8016    6758
## sbe3           0.28      0.46   -0.59    1.19 1.00    6374    5607
## sbe4           0.81      0.43   -0.02    1.68 1.00    7819    6003
## sze           0.24      0.40   -0.55    1.04 1.00    7223    6128
## som2           0.30      0.35   -0.38    1.00 1.00   10326    6234
## som3          -0.14      0.35   -0.84    0.54 1.00    8173    6191
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

S4.4 Learning rate updates

```
## Family: lognormal
## Links: mu = identity
## Formula: value ~ diagnosis * level * change + (level + change | subID)
## Data: df.upd (Number of observations: 176)
## Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
## total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~subID (Number of levels: 44)
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS
## sd(Intercept)      0.48      0.07    0.35    0.64 1.00   2751
## sd(level1)         0.74      0.09    0.57    0.94 1.00   2582
## sd(change1)        0.11      0.06    0.01    0.22 1.00   2400
## cor(Intercept,level1) 0.40      0.16    0.08    0.68 1.00   1321
## cor(Intercept,change1) 0.51      0.32   -0.31    0.92 1.00   4714
## cor(level1,change1)  0.15      0.33   -0.54    0.72 1.00   8925
##
##      Tail_ESS
## sd(Intercept)      4852
## sd(level1)         4058
## sd(change1)        1948
## cor(Intercept,level1) 2778
## cor(Intercept,change1) 4106
## cor(level1,change1)  5669
##
## Regression Coefficients:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS
## Intercept      -3.57      0.08   -3.74   -3.41 1.00   2132
## diagnosis1       0.17      0.09    0.00    0.33 1.00   2328
## level1          1.36      0.12    1.13    1.60 1.00   2131
## change1         1.11      0.05    1.02    1.20 1.00   7150
## diagnosis1:level1  0.03      0.12   -0.21    0.27 1.00   2149
## diagnosis1:change1 -0.09      0.05   -0.18   -0.00 1.00   7153
## level1:change1    -0.24      0.04   -0.32   -0.16 1.00  10154
## diagnosis1:level1:change1 0.12      0.04    0.04    0.20 1.00  10027
##
##      Tail_ESS
```



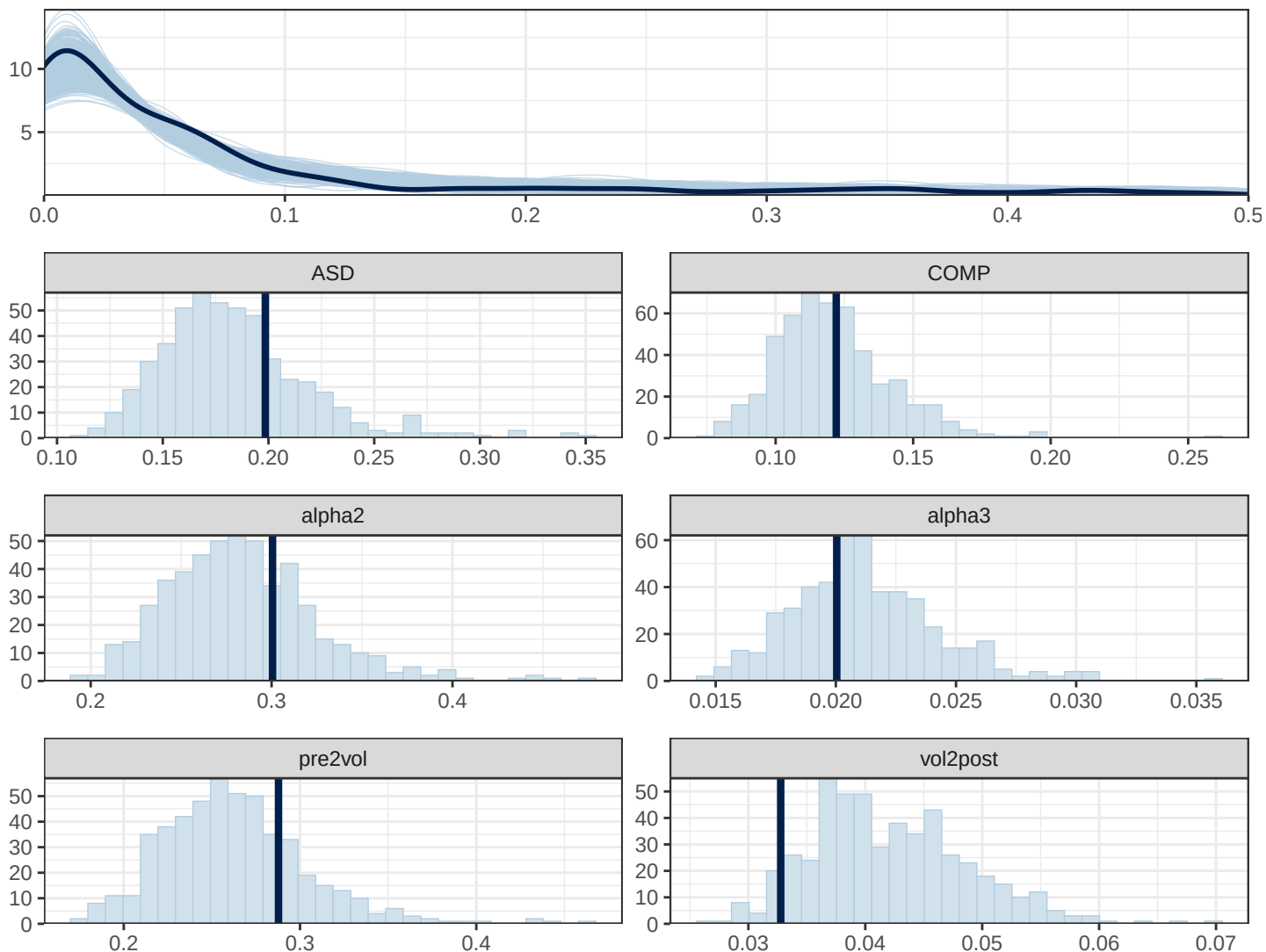
```
## Intercept                3428
## diagnosis1               3860
## level1                   3391
## change1                  5410
## diagnosis1:level1        3702
## diagnosis1:change1       5459
## level1:change1           6174
## diagnosis1:level1:change1 5466
##
## Further Distributional Parameters:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma      0.55      0.05   0.47   0.64 1.00   2647   4808
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

S4.4.1 Posterior predictive checks

```
## Warning: Removed 6814 rows containing non-finite outside the scale range
## (`stat_density()`).

## Warning: Removed 15 rows containing non-finite outside the scale range
## (`stat_density()`).
```

Posterior predictive checks



S4.4.2 ANOVA of ranked learning rate updates

```
## ---
## Model:
## Type: BFlinearModel, JZS
## Intercept only
## Data types:
## diagnosis : fixed
```

	bf
level + change	105.308214
level + change + level:change	105.274233
diagnosis + level + change + level:change	104.709253
diagnosis + level + change	104.630776
diagnosis + level + change + diagnosis:change	103.743090
diagnosis + level + change + diagnosis:change + level:change	103.632709
diagnosis + level + diagnosis:level + change + level:change	103.524108
diagnosis + level + diagnosis:level + change	103.470738
diagnosis + level + diagnosis:level + change + diagnosis:change + level:change	102.352071
diagnosis + level + diagnosis:level + change + diagnosis:change	102.350676
diagnosis + level + diagnosis:level + change + diagnosis:change + level:change + diagnosis:level:change	101.547006
level	51.218852
diagnosis + level	49.989893
diagnosis + level + diagnosis:level	48.643403
change	23.304343
diagnosis + change	21.885668
diagnosis + change + diagnosis:change	20.537285
diagnosis	-1.494272

S5 Reaction times, pupil sizes and accuracies

S5.1 Reaction times

```
## Family: shifted_lognormal
## Links: mu = identity
## Formula: rt.cor ~ diagnosis * expected * phase * difficulty + (expected * phase * difficulty | subID) + (di
## Data: df.pal (Number of observations: 11207)
## Draws: 4 chains, each with iter = 6000; warmup = 1500; thin = 1;
## total post-warmup draws = 18000
##
## Multilevel Hyperparameters:
## ~subID (Number of levels: 44)
##
## Estimate
## sd(Intercept) 0.16
## sd(expected1) 0.01
## sd(phase1) 0.06
## sd(phase2) 0.02
## sd(difficulty1) 0.01
## sd(difficulty2) 0.01
## sd(expected1:phase1) 0.01
## sd(expected1:phase2) 0.01
## sd(expected1:difficulty1) 0.00
## sd(expected1:difficulty2) 0.01
## sd(phase1:difficulty1) 0.01
## sd(phase2:difficulty1) 0.01
## sd(phase1:difficulty2) 0.01
## sd(phase2:difficulty2) 0.01
## sd(expected1:phase1:difficulty1) 0.01
## sd(expected1:phase2:difficulty1) 0.01
```

```

## sd(expected1:phase1:difficulty2) 0.01
## sd(expected1:phase2:difficulty2) 0.01
## cor(Intercept,expected1) -0.00
## cor(Intercept,phase1) -0.04
## cor(expected1,phase1) 0.12
## cor(Intercept,phase2) -0.01
## cor(expected1,phase2) -0.01
## cor(phase1,phase2) -0.14
## cor(Intercept,difficulty1) -0.04
## cor(expected1,difficulty1) 0.07
## cor(phase1,difficulty1) -0.16
## cor(phase2,difficulty1) 0.01
## cor(Intercept,difficulty2) -0.02
## cor(expected1,difficulty2) 0.03
## cor(phase1,difficulty2) -0.08
## cor(phase2,difficulty2) 0.01
## cor(difficulty1,difficulty2) 0.01
## cor(Intercept,expected1:phase1) -0.04
## cor(expected1,expected1:phase1) 0.05
## cor(phase1,expected1:phase1) 0.04
## cor(phase2,expected1:phase1) -0.08
## cor(difficulty1,expected1:phase1) -0.01
## cor(difficulty2,expected1:phase1) 0.00
## cor(Intercept,expected1:phase2) 0.05
## cor(expected1,expected1:phase2) -0.01
## cor(phase1,expected1:phase2) -0.08
## cor(phase2,expected1:phase2) -0.06
## cor(difficulty1,expected1:phase2) 0.03
## cor(difficulty2,expected1:phase2) 0.06
## cor(expected1:phase1,expected1:phase2) -0.04
## cor(Intercept,expected1:difficulty1) -0.04
## cor(expected1,expected1:difficulty1) 0.02
## cor(phase1,expected1:difficulty1) 0.01
## cor(phase2,expected1:difficulty1) 0.05
## cor(difficulty1,expected1:difficulty1) -0.03
## cor(difficulty2,expected1:difficulty1) -0.00
## cor(expected1:phase1,expected1:difficulty1) 0.00
## cor(expected1:phase2,expected1:difficulty1) -0.01
## cor(Intercept,expected1:difficulty2) -0.07
## cor(expected1,expected1:difficulty2) 0.04
## cor(phase1,expected1:difficulty2) 0.29
## cor(phase2,expected1:difficulty2) 0.01
## cor(difficulty1,expected1:difficulty2) -0.09
## cor(difficulty2,expected1:difficulty2) -0.06
## cor(expected1:phase1,expected1:difficulty2) -0.00
## cor(expected1:phase2,expected1:difficulty2) 0.04
## cor(expected1:difficulty1,expected1:difficulty2) -0.04
## cor(Intercept,phase1:difficulty1) -0.02
## cor(expected1,phase1:difficulty1) -0.03
## cor(phase1,phase1:difficulty1) 0.02
## cor(phase2,phase1:difficulty1) 0.02
## cor(difficulty1,phase1:difficulty1) -0.01
## cor(difficulty2,phase1:difficulty1) -0.00
## cor(expected1:phase1,phase1:difficulty1) -0.02
## cor(expected1:phase2,phase1:difficulty1) -0.01
## cor(expected1:difficulty1,phase1:difficulty1) 0.01
## cor(expected1:difficulty2,phase1:difficulty1) 0.04
## cor(Intercept,phase2:difficulty1) -0.03
## cor(expected1,phase2:difficulty1) 0.10
## cor(phase1,phase2:difficulty1) -0.08

```

```

## cor(phase2,phase2:difficulty1) -0.05
## cor(difficulty1,phase2:difficulty1) 0.09
## cor(difficulty2,phase2:difficulty1) 0.05
## cor(expected1:phase1,phase2:difficulty1) -0.02
## cor(expected1:phase2,phase2:difficulty1) 0.02
## cor(expected1:difficulty1,phase2:difficulty1) 0.02
## cor(expected1:difficulty2,phase2:difficulty1) 0.02
## cor(phase1:difficulty1,phase2:difficulty1) -0.04
## cor(Intercept,phase1:difficulty2) 0.03
## cor(expected1,phase1:difficulty2) -0.02
## cor(phase1,phase1:difficulty2) 0.04
## cor(phase2,phase1:difficulty2) 0.05
## cor(difficulty1,phase1:difficulty2) -0.04
## cor(difficulty2,phase1:difficulty2) 0.00
## cor(expected1:phase1,phase1:difficulty2) -0.03
## cor(expected1:phase2,phase1:difficulty2) -0.01
## cor(expected1:difficulty1,phase1:difficulty2) -0.01
## cor(expected1:difficulty2,phase1:difficulty2) 0.06
## cor(phase1:difficulty1,phase1:difficulty2) -0.02
## cor(phase2:difficulty1,phase1:difficulty2) -0.03
## cor(Intercept,phase2:difficulty2) 0.04
## cor(expected1,phase2:difficulty2) 0.01
## cor(phase1,phase2:difficulty2) -0.08
## cor(phase2,phase2:difficulty2) -0.01
## cor(difficulty1,phase2:difficulty2) 0.02
## cor(difficulty2,phase2:difficulty2) 0.01
## cor(expected1:phase1,phase2:difficulty2) 0.02
## cor(expected1:phase2,phase2:difficulty2) 0.02
## cor(expected1:difficulty1,phase2:difficulty2) -0.01
## cor(expected1:difficulty2,phase2:difficulty2) -0.04
## cor(phase1:difficulty1,phase2:difficulty2) -0.03
## cor(phase2:difficulty1,phase2:difficulty2) -0.05
## cor(phase1:difficulty2,phase2:difficulty2) -0.01
## cor(Intercept,expected1:phase1:difficulty1) 0.04
## cor(expected1,expected1:phase1:difficulty1) -0.04
## cor(phase1,expected1:phase1:difficulty1) -0.01
## cor(phase2,expected1:phase1:difficulty1) 0.05
## cor(difficulty1,expected1:phase1:difficulty1) -0.00
## cor(difficulty2,expected1:phase1:difficulty1) -0.01
## cor(expected1:phase1,expected1:phase1:difficulty1) -0.01
## cor(expected1:phase2,expected1:phase1:difficulty1) 0.01
## cor(expected1:difficulty1,expected1:phase1:difficulty1) 0.01
## cor(expected1:difficulty2,expected1:phase1:difficulty1) 0.00
## cor(phase1:difficulty1,expected1:phase1:difficulty1) -0.02
## cor(phase2:difficulty1,expected1:phase1:difficulty1) -0.03
## cor(phase1:difficulty2,expected1:phase1:difficulty1) -0.02
## cor(phase2:difficulty2,expected1:phase1:difficulty1) -0.02
## cor(Intercept,expected1:phase2:difficulty1) 0.10
## cor(expected1,expected1:phase2:difficulty1) 0.02
## cor(phase1,expected1:phase2:difficulty1) 0.04
## cor(phase2,expected1:phase2:difficulty1) 0.07
## cor(difficulty1,expected1:phase2:difficulty1) -0.00
## cor(difficulty2,expected1:phase2:difficulty1) 0.00
## cor(expected1:phase1,expected1:phase2:difficulty1) -0.00
## cor(expected1:phase2,expected1:phase2:difficulty1) 0.01
## cor(expected1:difficulty1,expected1:phase2:difficulty1) -0.00
## cor(expected1:difficulty2,expected1:phase2:difficulty1) 0.00
## cor(phase1:difficulty1,expected1:phase2:difficulty1) -0.01
## cor(phase2:difficulty1,expected1:phase2:difficulty1) -0.04
## cor(phase1:difficulty2,expected1:phase2:difficulty1) 0.00

```

## cor(phase2:difficulty2,expected1:phase2:difficulty1)	-0.02
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty1)	-0.01
## cor(Intercept,expected1:phase1:difficulty2)	0.04
## cor(expected1,expected1:phase1:difficulty2)	-0.06
## cor(phase1,expected1:phase1:difficulty2)	-0.06
## cor(phase2,expected1:phase1:difficulty2)	0.10
## cor(difficulty1,expected1:phase1:difficulty2)	-0.01
## cor(difficulty2,expected1:phase1:difficulty2)	0.03
## cor(expected1:phase1,expected1:phase1:difficulty2)	-0.05
## cor(expected1:phase2,expected1:phase1:difficulty2)	0.03
## cor(expected1:difficulty1,expected1:phase1:difficulty2)	-0.01
## cor(expected1:difficulty2,expected1:phase1:difficulty2)	0.04
## cor(phase1:difficulty1,expected1:phase1:difficulty2)	-0.01
## cor(phase2:difficulty1,expected1:phase1:difficulty2)	-0.02
## cor(phase1:difficulty2,expected1:phase1:difficulty2)	-0.02
## cor(phase2:difficulty2,expected1:phase1:difficulty2)	-0.00
## cor(expected1:phase1:difficulty1,expected1:phase1:difficulty2)	-0.01
## cor(expected1:phase2:difficulty1,expected1:phase1:difficulty2)	0.01
## cor(Intercept,expected1:phase2:difficulty2)	0.09
## cor(expected1,expected1:phase2:difficulty2)	-0.01
## cor(phase1,expected1:phase2:difficulty2)	-0.04
## cor(phase2,expected1:phase2:difficulty2)	0.08
## cor(difficulty1,expected1:phase2:difficulty2)	0.00
## cor(difficulty2,expected1:phase2:difficulty2)	0.01
## cor(expected1:phase1,expected1:phase2:difficulty2)	0.00
## cor(expected1:phase2,expected1:phase2:difficulty2)	0.03
## cor(expected1:difficulty1,expected1:phase2:difficulty2)	-0.02
## cor(expected1:difficulty2,expected1:phase2:difficulty2)	-0.03
## cor(phase1:difficulty1,expected1:phase2:difficulty2)	-0.00
## cor(phase2:difficulty1,expected1:phase2:difficulty2)	-0.04
## cor(phase1:difficulty2,expected1:phase2:difficulty2)	0.00
## cor(phase2:difficulty2,expected1:phase2:difficulty2)	-0.02
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty2)	0.01
## cor(expected1:phase2:difficulty1,expected1:phase2:difficulty2)	-0.02
## cor(expected1:phase1:difficulty2,expected1:phase2:difficulty2)	0.01
##	Est.Error
## sd(Intercept)	0.02
## sd(expected1)	0.01
## sd(phase1)	0.01
## sd(phase2)	0.00
## sd(difficulty1)	0.00
## sd(difficulty2)	0.00
## sd(expected1:phase1)	0.01
## sd(expected1:phase2)	0.00
## sd(expected1:difficulty1)	0.00
## sd(expected1:difficulty2)	0.00
## sd(phase1:difficulty1)	0.00
## sd(phase2:difficulty1)	0.01
## sd(phase1:difficulty2)	0.00
## sd(phase2:difficulty2)	0.01
## sd(expected1:phase1:difficulty1)	0.00
## sd(expected1:phase2:difficulty1)	0.00
## sd(expected1:phase1:difficulty2)	0.01
## sd(expected1:phase2:difficulty2)	0.00
## cor(Intercept,expected1)	0.18
## cor(Intercept,phase1)	0.13
## cor(expected1,phase1)	0.18
## cor(Intercept,phase2)	0.15
## cor(expected1,phase2)	0.19
## cor(phase1,phase2)	0.16

```

## cor(Intercept,difficulty1) 0.20
## cor(expected1,difficulty1) 0.21
## cor(phase1,difficulty1) 0.21
## cor(phase2,difficulty1) 0.21
## cor(Intercept,difficulty2) 0.21
## cor(expected1,difficulty2) 0.21
## cor(phase1,difficulty2) 0.21
## cor(phase2,difficulty2) 0.21
## cor(difficulty1,difficulty2) 0.22
## cor(Intercept,expected1:phase1) 0.20
## cor(expected1,expected1:phase1) 0.22
## cor(phase1,expected1:phase1) 0.20
## cor(phase2,expected1:phase1) 0.21
## cor(difficulty1,expected1:phase1) 0.22
## cor(difficulty2,expected1:phase1) 0.22
## cor(Intercept,expected1:phase2) 0.21
## cor(expected1,expected1:phase2) 0.22
## cor(phase1,expected1:phase2) 0.21
## cor(phase2,expected1:phase2) 0.22
## cor(difficulty1,expected1:phase2) 0.22
## cor(difficulty2,expected1:phase2) 0.22
## cor(expected1:phase1,expected1:phase2) 0.22
## cor(Intercept,expected1:difficulty1) 0.21
## cor(expected1,expected1:difficulty1) 0.22
## cor(phase1,expected1:difficulty1) 0.21
## cor(phase2,expected1:difficulty1) 0.21
## cor(difficulty1,expected1:difficulty1) 0.22
## cor(difficulty2,expected1:difficulty1) 0.22
## cor(expected1:phase1,expected1:difficulty1) 0.22
## cor(expected1:phase2,expected1:difficulty1) 0.22
## cor(Intercept,expected1:difficulty2) 0.17
## cor(expected1,expected1:difficulty2) 0.20
## cor(phase1,expected1:difficulty2) 0.17
## cor(phase2,expected1:difficulty2) 0.19
## cor(difficulty1,expected1:difficulty2) 0.22
## cor(difficulty2,expected1:difficulty2) 0.22
## cor(expected1:phase1,expected1:difficulty2) 0.21
## cor(expected1:phase2,expected1:difficulty2) 0.22
## cor(expected1:difficulty1,expected1:difficulty2) 0.22
## cor(Intercept,phase1:difficulty1) 0.21
## cor(expected1,phase1:difficulty1) 0.22
## cor(phase1,phase1:difficulty1) 0.21
## cor(phase2,phase1:difficulty1) 0.22
## cor(difficulty1,phase1:difficulty1) 0.22
## cor(difficulty2,phase1:difficulty1) 0.22
## cor(expected1:phase1,phase1:difficulty1) 0.22
## cor(expected1:phase2,phase1:difficulty1) 0.22
## cor(expected1:difficulty1,phase1:difficulty1) 0.22
## cor(expected1:difficulty2,phase1:difficulty1) 0.22
## cor(Intercept,phase2:difficulty1) 0.19
## cor(expected1,phase2:difficulty1) 0.20
## cor(phase1,phase2:difficulty1) 0.19
## cor(phase2,phase2:difficulty1) 0.20
## cor(difficulty1,phase2:difficulty1) 0.21
## cor(difficulty2,phase2:difficulty1) 0.22
## cor(expected1:phase1,phase2:difficulty1) 0.22
## cor(expected1:phase2,phase2:difficulty1) 0.22
## cor(expected1:difficulty1,phase2:difficulty1) 0.22
## cor(expected1:difficulty2,phase2:difficulty1) 0.20
## cor(phase1:difficulty1,phase2:difficulty1) 0.22

```

```

## cor(Intercept,phase1:difficulty2) 0.21
## cor(expected1,phase1:difficulty2) 0.22
## cor(phase1,phase1:difficulty2) 0.22
## cor(phase2,phase1:difficulty2) 0.22
## cor(difficulty1,phase1:difficulty2) 0.22
## cor(difficulty2,phase1:difficulty2) 0.22
## cor(expected1:phase1,phase1:difficulty2) 0.22
## cor(expected1:phase2,phase1:difficulty2) 0.22
## cor(expected1:difficulty1,phase1:difficulty2) 0.22
## cor(expected1:difficulty2,phase1:difficulty2) 0.21
## cor(phase1:difficulty1,phase1:difficulty2) 0.22
## cor(phase2:difficulty1,phase1:difficulty2) 0.22
## cor(Intercept,phase2:difficulty2) 0.20
## cor(expected1,phase2:difficulty2) 0.21
## cor(phase1,phase2:difficulty2) 0.20
## cor(phase2,phase2:difficulty2) 0.21
## cor(difficulty1,phase2:difficulty2) 0.21
## cor(difficulty2,phase2:difficulty2) 0.22
## cor(expected1:phase1,phase2:difficulty2) 0.22
## cor(expected1:phase2,phase2:difficulty2) 0.22
## cor(expected1:difficulty1,phase2:difficulty2) 0.22
## cor(expected1:difficulty2,phase2:difficulty2) 0.21
## cor(phase1:difficulty1,phase2:difficulty2) 0.22
## cor(phase2:difficulty1,phase2:difficulty2) 0.22
## cor(phase1:difficulty2,phase2:difficulty2) 0.22
## cor(Intercept,expected1:phase1:difficulty1) 0.21
## cor(expected1,expected1:phase1:difficulty1) 0.21
## cor(phase1,expected1:phase1:difficulty1) 0.21
## cor(phase2,expected1:phase1:difficulty1) 0.22
## cor(difficulty1,expected1:phase1:difficulty1) 0.22
## cor(difficulty2,expected1:phase1:difficulty1) 0.22
## cor(expected1:phase1,expected1:phase1:difficulty1) 0.22
## cor(expected1:phase2,expected1:phase1:difficulty1) 0.22
## cor(expected1:difficulty1,expected1:phase1:difficulty1) 0.22
## cor(expected1:difficulty2,expected1:phase1:difficulty1) 0.21
## cor(phase1:difficulty1,expected1:phase1:difficulty1) 0.22
## cor(phase2:difficulty1,expected1:phase1:difficulty1) 0.22
## cor(phase1:difficulty2,expected1:phase1:difficulty1) 0.22
## cor(phase2:difficulty2,expected1:phase1:difficulty1) 0.22
## cor(Intercept,expected1:phase2:difficulty1) 0.21
## cor(expected1,expected1:phase2:difficulty1) 0.22
## cor(phase1,expected1:phase2:difficulty1) 0.21
## cor(phase2,expected1:phase2:difficulty1) 0.22
## cor(difficulty1,expected1:phase2:difficulty1) 0.22
## cor(difficulty2,expected1:phase2:difficulty1) 0.22
## cor(expected1:phase1,expected1:phase2:difficulty1) 0.22
## cor(expected1:phase2,expected1:phase2:difficulty1) 0.22
## cor(expected1:difficulty1,expected1:phase2:difficulty1) 0.22
## cor(expected1:difficulty2,expected1:phase2:difficulty1) 0.21
## cor(phase1:difficulty1,expected1:phase2:difficulty1) 0.22
## cor(phase2:difficulty1,expected1:phase2:difficulty1) 0.22
## cor(phase1:difficulty2,expected1:phase2:difficulty1) 0.22
## cor(phase2:difficulty2,expected1:phase2:difficulty1) 0.22
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty1) 0.22
## cor(Intercept,expected1:phase1:difficulty2) 0.21
## cor(expected1,expected1:phase1:difficulty2) 0.22
## cor(phase1,expected1:phase1:difficulty2) 0.21
## cor(phase2,expected1:phase1:difficulty2) 0.21
## cor(difficulty1,expected1:phase1:difficulty2) 0.22
## cor(difficulty2,expected1:phase1:difficulty2) 0.22

```

```

## cor(expected1:phase1,expected1:phase1:difficulty2) 0.22
## cor(expected1:phase2,expected1:phase1:difficulty2) 0.22
## cor(expected1:difficulty1,expected1:phase1:difficulty2) 0.22
## cor(expected1:difficulty2,expected1:phase1:difficulty2) 0.21
## cor(phase1:difficulty1,expected1:phase1:difficulty2) 0.22
## cor(phase2:difficulty1,expected1:phase1:difficulty2) 0.22
## cor(phase1:difficulty2,expected1:phase1:difficulty2) 0.22
## cor(phase2:difficulty2,expected1:phase1:difficulty2) 0.22
## cor(expected1:phase1:difficulty1,expected1:phase1:difficulty2) 0.22
## cor(expected1:phase2:difficulty1,expected1:phase1:difficulty2) 0.22
## cor(Intercept,expected1:phase2:difficulty2) 0.22
## cor(expected1,expected1:phase2:difficulty2) 0.22
## cor(phase1,expected1:phase2:difficulty2) 0.21
## cor(phase2,expected1:phase2:difficulty2) 0.22
## cor(difficulty1,expected1:phase2:difficulty2) 0.22
## cor(difficulty2,expected1:phase2:difficulty2) 0.22
## cor(expected1:phase1,expected1:phase2:difficulty2) 0.22
## cor(expected1:phase2,expected1:phase2:difficulty2) 0.22
## cor(expected1:difficulty1,expected1:phase2:difficulty2) 0.22
## cor(expected1:difficulty2,expected1:phase2:difficulty2) 0.22
## cor(phase1:difficulty1,expected1:phase2:difficulty2) 0.22
## cor(phase2:difficulty1,expected1:phase2:difficulty2) 0.22
## cor(phase1:difficulty2,expected1:phase2:difficulty2) 0.22
## cor(phase2:difficulty2,expected1:phase2:difficulty2) 0.22
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty2) 0.22
## cor(expected1:phase2:difficulty1,expected1:phase2:difficulty2) 0.22
## cor(expected1:phase1:difficulty2,expected1:phase2:difficulty2) 0.22
## 1-95% CI
## sd(Intercept) 0.13
## sd(expected1) 0.00
## sd(phase1) 0.04
## sd(phase2) 0.01
## sd(difficulty1) 0.00
## sd(difficulty2) 0.00
## sd(expected1:phase1) 0.00
## sd(expected1:phase2) 0.00
## sd(expected1:difficulty1) 0.00
## sd(expected1:difficulty2) 0.00
## sd(phase1:difficulty1) 0.00
## sd(phase2:difficulty1) 0.00
## sd(phase1:difficulty2) 0.00
## sd(phase2:difficulty2) 0.00
## sd(expected1:phase1:difficulty1) 0.00
## sd(expected1:phase2:difficulty1) 0.00
## sd(expected1:phase1:difficulty2) 0.00
## sd(expected1:phase2:difficulty2) 0.00
## cor(Intercept,expected1) -0.34
## cor(Intercept,phase1) -0.30
## cor(expected1,phase1) -0.24
## cor(Intercept,phase2) -0.31
## cor(expected1,phase2) -0.39
## cor(phase1,phase2) -0.45
## cor(Intercept,difficulty1) -0.42
## cor(expected1,difficulty1) -0.36
## cor(phase1,difficulty1) -0.54
## cor(phase2,difficulty1) -0.40
## cor(Intercept,difficulty2) -0.42
## cor(expected1,difficulty2) -0.38
## cor(phase1,difficulty2) -0.48
## cor(phase2,difficulty2) -0.40

```



```

## cor(difficulty1,difficulty2) -0.41
## cor(Intercept,expected1:phase1) -0.42
## cor(expected1,expected1:phase1) -0.38
## cor(phase1,expected1:phase1) -0.36
## cor(phase2,expected1:phase1) -0.47
## cor(difficulty1,expected1:phase1) -0.42
## cor(difficulty2,expected1:phase1) -0.42
## cor(Intercept,expected1:phase2) -0.36
## cor(expected1,expected1:phase2) -0.43
## cor(phase1,expected1:phase2) -0.48
## cor(phase2,expected1:phase2) -0.48
## cor(difficulty1,expected1:phase2) -0.39
## cor(difficulty2,expected1:phase2) -0.38
## cor(expected1:phase1,expected1:phase2) -0.47
## cor(Intercept,expected1:difficulty1) -0.44
## cor(expected1,expected1:difficulty1) -0.41
## cor(phase1,expected1:difficulty1) -0.41
## cor(phase2,expected1:difficulty1) -0.37
## cor(difficulty1,expected1:difficulty1) -0.46
## cor(difficulty2,expected1:difficulty1) -0.43
## cor(expected1:phase1,expected1:difficulty1) -0.42
## cor(expected1:phase2,expected1:difficulty1) -0.43
## cor(Intercept,expected1:difficulty2) -0.40
## cor(expected1,expected1:difficulty2) -0.35
## cor(phase1,expected1:difficulty2) -0.08
## cor(phase2,expected1:difficulty2) -0.36
## cor(difficulty1,expected1:difficulty2) -0.49
## cor(difficulty2,expected1:difficulty2) -0.47
## cor(expected1:phase1,expected1:difficulty2) -0.41
## cor(expected1:phase2,expected1:difficulty2) -0.38
## cor(expected1:difficulty1,expected1:difficulty2) -0.46
## cor(Intercept,phase1:difficulty1) -0.43
## cor(expected1,phase1:difficulty1) -0.45
## cor(phase1,phase1:difficulty1) -0.40
## cor(phase2,phase1:difficulty1) -0.41
## cor(difficulty1,phase1:difficulty1) -0.43
## cor(difficulty2,phase1:difficulty1) -0.43
## cor(expected1:phase1,phase1:difficulty1) -0.44
## cor(expected1:phase2,phase1:difficulty1) -0.43
## cor(expected1:difficulty1,phase1:difficulty1) -0.42
## cor(expected1:difficulty2,phase1:difficulty1) -0.39
## cor(Intercept,phase2:difficulty1) -0.39
## cor(expected1,phase2:difficulty1) -0.31
## cor(phase1,phase2:difficulty1) -0.43
## cor(phase2,phase2:difficulty1) -0.43
## cor(difficulty1,phase2:difficulty1) -0.33
## cor(difficulty2,phase2:difficulty1) -0.38
## cor(expected1:phase1,phase2:difficulty1) -0.43
## cor(expected1:phase2,phase2:difficulty1) -0.40
## cor(expected1:difficulty1,phase2:difficulty1) -0.40
## cor(expected1:difficulty2,phase2:difficulty1) -0.38
## cor(phase1:difficulty1,phase2:difficulty1) -0.46
## cor(Intercept,phase1:difficulty2) -0.39
## cor(expected1,phase1:difficulty2) -0.44
## cor(phase1,phase1:difficulty2) -0.38
## cor(phase2,phase1:difficulty2) -0.38
## cor(difficulty1,phase1:difficulty2) -0.45
## cor(difficulty2,phase1:difficulty2) -0.42
## cor(expected1:phase1,phase1:difficulty2) -0.45
## cor(expected1:phase2,phase1:difficulty2) -0.43

```

```

## cor(expected1:difficulty1,phase1:difficulty2) -0.43
## cor(expected1:difficulty2,phase1:difficulty2) -0.36
## cor(phase1:difficulty1,phase1:difficulty2) -0.44
## cor(phase2:difficulty1,phase1:difficulty2) -0.45
## cor(Intercept,phase2:difficulty2) -0.35
## cor(expected1,phase2:difficulty2) -0.41
## cor(phase1,phase2:difficulty2) -0.47
## cor(phase2,phase2:difficulty2) -0.40
## cor(difficulty1,phase2:difficulty2) -0.39
## cor(difficulty2,phase2:difficulty2) -0.41
## cor(expected1:phase1,phase2:difficulty2) -0.41
## cor(expected1:phase2,phase2:difficulty2) -0.40
## cor(expected1:difficulty1,phase2:difficulty2) -0.44
## cor(expected1:difficulty2,phase2:difficulty2) -0.44
## cor(phase1:difficulty1,phase2:difficulty2) -0.45
## cor(phase2:difficulty1,phase2:difficulty2) -0.46
## cor(phase1:difficulty2,phase2:difficulty2) -0.43
## cor(Intercept,expected1:phase1:difficulty1) -0.37
## cor(expected1,expected1:phase1:difficulty1) -0.45
## cor(phase1,expected1:phase1:difficulty1) -0.42
## cor(phase2,expected1:phase1:difficulty1) -0.38
## cor(difficulty1,expected1:phase1:difficulty1) -0.42
## cor(difficulty2,expected1:phase1:difficulty1) -0.44
## cor(expected1:phase1,expected1:phase1:difficulty1) -0.43
## cor(expected1:phase2,expected1:phase1:difficulty1) -0.42
## cor(expected1:difficulty1,expected1:phase1:difficulty1) -0.41
## cor(expected1:difficulty2,expected1:phase1:difficulty1) -0.41
## cor(phase1:difficulty1,expected1:phase1:difficulty1) -0.45
## cor(phase2:difficulty1,expected1:phase1:difficulty1) -0.45
## cor(phase1:difficulty2,expected1:phase1:difficulty1) -0.44
## cor(phase2:difficulty2,expected1:phase1:difficulty1) -0.44
## cor(Intercept,expected1:phase2:difficulty1) -0.33
## cor(expected1,expected1:phase2:difficulty1) -0.40
## cor(phase1,expected1:phase2:difficulty1) -0.38
## cor(phase2,expected1:phase2:difficulty1) -0.35
## cor(difficulty1,expected1:phase2:difficulty1) -0.42
## cor(difficulty2,expected1:phase2:difficulty1) -0.41
## cor(expected1:phase1,expected1:phase2:difficulty1) -0.42
## cor(expected1:phase2,expected1:phase2:difficulty1) -0.42
## cor(expected1:difficulty1,expected1:phase2:difficulty1) -0.43
## cor(expected1:difficulty2,expected1:phase2:difficulty1) -0.41
## cor(phase1:difficulty1,expected1:phase2:difficulty1) -0.43
## cor(phase2:difficulty1,expected1:phase2:difficulty1) -0.45
## cor(phase1:difficulty2,expected1:phase2:difficulty1) -0.42
## cor(phase2:difficulty2,expected1:phase2:difficulty1) -0.44
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty1) -0.43
## cor(Intercept,expected1:phase1:difficulty2) -0.37
## cor(expected1,expected1:phase1:difficulty2) -0.47
## cor(phase1,expected1:phase1:difficulty2) -0.44
## cor(phase2,expected1:phase1:difficulty2) -0.33
## cor(difficulty1,expected1:phase1:difficulty2) -0.43
## cor(difficulty2,expected1:phase1:difficulty2) -0.40
## cor(expected1:phase1,expected1:phase1:difficulty2) -0.46
## cor(expected1:phase2,expected1:phase1:difficulty2) -0.39
## cor(expected1:difficulty1,expected1:phase1:difficulty2) -0.43
## cor(expected1:difficulty2,expected1:phase1:difficulty2) -0.38
## cor(phase1:difficulty1,expected1:phase1:difficulty2) -0.43
## cor(phase2:difficulty1,expected1:phase1:difficulty2) -0.43
## cor(phase1:difficulty2,expected1:phase1:difficulty2) -0.44
## cor(phase2:difficulty2,expected1:phase1:difficulty2) -0.42

```

```

## cor(expected1:phase1:difficulty1,expected1:phase1:difficulty2) -0.43
## cor(expected1:phase2:difficulty1,expected1:phase1:difficulty2) -0.41
## cor(Intercept,expected1:phase2:difficulty2) -0.34
## cor(expected1,expected1:phase2:difficulty2) -0.43
## cor(phase1,expected1:phase2:difficulty2) -0.45
## cor(phase2,expected1:phase2:difficulty2) -0.35
## cor(difficulty1,expected1:phase2:difficulty2) -0.42
## cor(difficulty2,expected1:phase2:difficulty2) -0.41
## cor(expected1:phase1,expected1:phase2:difficulty2) -0.42
## cor(expected1:phase2,expected1:phase2:difficulty2) -0.40
## cor(expected1:difficulty1,expected1:phase2:difficulty2) -0.44
## cor(expected1:difficulty2,expected1:phase2:difficulty2) -0.45
## cor(phase1:difficulty1,expected1:phase2:difficulty2) -0.42
## cor(phase2:difficulty1,expected1:phase2:difficulty2) -0.45
## cor(phase1:difficulty2,expected1:phase2:difficulty2) -0.41
## cor(phase2:difficulty2,expected1:phase2:difficulty2) -0.43
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty2) -0.41
## cor(expected1:phase2:difficulty1,expected1:phase2:difficulty2) -0.44
## cor(expected1:phase1:difficulty2,expected1:phase2:difficulty2) -0.42
##
## u-95% CI Rhat
## sd(Intercept) 0.20 1.00
## sd(expected1) 0.02 1.00
## sd(phase1) 0.07 1.00
## sd(phase2) 0.03 1.00
## sd(difficulty1) 0.02 1.00
## sd(difficulty2) 0.02 1.00
## sd(expected1:phase1) 0.02 1.00
## sd(expected1:phase2) 0.02 1.00
## sd(expected1:difficulty1) 0.01 1.00
## sd(expected1:difficulty2) 0.02 1.00
## sd(phase1:difficulty1) 0.02 1.00
## sd(phase2:difficulty1) 0.03 1.00
## sd(phase1:difficulty2) 0.02 1.00
## sd(phase2:difficulty2) 0.02 1.00
## sd(expected1:phase1:difficulty1) 0.02 1.00
## sd(expected1:phase2:difficulty1) 0.02 1.00
## sd(expected1:phase1:difficulty2) 0.02 1.00
## sd(expected1:phase2:difficulty2) 0.02 1.00
## cor(Intercept,expected1) 0.35 1.00
## cor(Intercept,phase1) 0.22 1.00
## cor(expected1,phase1) 0.46 1.00
## cor(Intercept,phase2) 0.29 1.00
## cor(expected1,phase2) 0.37 1.00
## cor(phase1,phase2) 0.18 1.00
## cor(Intercept,difficulty1) 0.35 1.00
## cor(expected1,difficulty1) 0.46 1.00
## cor(phase1,difficulty1) 0.26 1.00
## cor(phase2,difficulty1) 0.41 1.00
## cor(Intercept,difficulty2) 0.39 1.00
## cor(expected1,difficulty2) 0.44 1.00
## cor(phase1,difficulty2) 0.34 1.00
## cor(phase2,difficulty2) 0.42 1.00
## cor(difficulty1,difficulty2) 0.43 1.00
## cor(Intercept,expected1:phase1) 0.36 1.00
## cor(expected1,expected1:phase1) 0.47 1.00
## cor(phase1,expected1:phase1) 0.43 1.00
## cor(phase2,expected1:phase1) 0.35 1.00
## cor(difficulty1,expected1:phase1) 0.41 1.00
## cor(difficulty2,expected1:phase1) 0.42 1.00
## cor(Intercept,expected1:phase2) 0.45 1.00

```

```

## cor(expected1,expected1:phase2) 0.42 1.00
## cor(phase1,expected1:phase2) 0.34 1.00
## cor(phase2,expected1:phase2) 0.36 1.00
## cor(difficulty1,expected1:phase2) 0.45 1.00
## cor(difficulty2,expected1:phase2) 0.48 1.00
## cor(expected1:phase1,expected1:phase2) 0.38 1.00
## cor(Intercept,expected1:difficulty1) 0.38 1.00
## cor(expected1,expected1:difficulty1) 0.44 1.00
## cor(phase1,expected1:difficulty1) 0.41 1.00
## cor(phase2,expected1:difficulty1) 0.46 1.00
## cor(difficulty1,expected1:difficulty1) 0.41 1.00
## cor(difficulty2,expected1:difficulty1) 0.42 1.00
## cor(expected1:phase1,expected1:difficulty1) 0.42 1.00
## cor(expected1:phase2,expected1:difficulty1) 0.41 1.00
## cor(Intercept,expected1:difficulty2) 0.27 1.00
## cor(expected1,expected1:difficulty2) 0.43 1.00
## cor(phase1,expected1:difficulty2) 0.60 1.00
## cor(phase2,expected1:difficulty2) 0.38 1.00
## cor(difficulty1,expected1:difficulty2) 0.34 1.00
## cor(difficulty2,expected1:difficulty2) 0.37 1.00
## cor(expected1:phase1,expected1:difficulty2) 0.41 1.00
## cor(expected1:phase2,expected1:difficulty2) 0.45 1.00
## cor(expected1:difficulty1,expected1:difficulty2) 0.39 1.00
## cor(Intercept,phase1:difficulty1) 0.39 1.00
## cor(expected1,phase1:difficulty1) 0.39 1.00
## cor(phase1,phase1:difficulty1) 0.43 1.00
## cor(phase2,phase1:difficulty1) 0.43 1.00
## cor(difficulty1,phase1:difficulty1) 0.42 1.00
## cor(difficulty2,phase1:difficulty1) 0.42 1.00
## cor(expected1:phase1,phase1:difficulty1) 0.40 1.00
## cor(expected1:phase2,phase1:difficulty1) 0.41 1.00
## cor(expected1:difficulty1,phase1:difficulty1) 0.43 1.00
## cor(expected1:difficulty2,phase1:difficulty1) 0.45 1.00
## cor(Intercept,phase2:difficulty1) 0.33 1.00
## cor(expected1,phase2:difficulty1) 0.48 1.00
## cor(phase1,phase2:difficulty1) 0.29 1.00
## cor(phase2,phase2:difficulty1) 0.34 1.00
## cor(difficulty1,phase2:difficulty1) 0.49 1.00
## cor(difficulty2,phase2:difficulty1) 0.46 1.00
## cor(expected1:phase1,phase2:difficulty1) 0.40 1.00
## cor(expected1:phase2,phase2:difficulty1) 0.44 1.00
## cor(expected1:difficulty1,phase2:difficulty1) 0.44 1.00
## cor(expected1:difficulty2,phase2:difficulty1) 0.42 1.00
## cor(phase1:difficulty1,phase2:difficulty1) 0.39 1.00
## cor(Intercept,phase1:difficulty2) 0.44 1.00
## cor(expected1,phase1:difficulty2) 0.40 1.00
## cor(phase1,phase1:difficulty2) 0.45 1.00
## cor(phase2,phase1:difficulty2) 0.47 1.00
## cor(difficulty1,phase1:difficulty2) 0.39 1.00
## cor(difficulty2,phase1:difficulty2) 0.42 1.00
## cor(expected1:phase1,phase1:difficulty2) 0.40 1.00
## cor(expected1:phase2,phase1:difficulty2) 0.41 1.00
## cor(expected1:difficulty1,phase1:difficulty2) 0.42 1.00
## cor(expected1:difficulty2,phase1:difficulty2) 0.46 1.00
## cor(phase1:difficulty1,phase1:difficulty2) 0.42 1.00
## cor(phase2:difficulty1,phase1:difficulty2) 0.39 1.00
## cor(Intercept,phase2:difficulty2) 0.42 1.00
## cor(expected1,phase2:difficulty2) 0.42 1.00
## cor(phase1,phase2:difficulty2) 0.33 1.00
## cor(phase2,phase2:difficulty2) 0.40 1.00

```

```

## cor(difficulty1,phase2:difficulty2) 0.44 1.00
## cor(difficulty2,phase2:difficulty2) 0.43 1.00
## cor(expected1:phase1,phase2:difficulty2) 0.44 1.00
## cor(expected1:phase2,phase2:difficulty2) 0.43 1.00
## cor(expected1:difficulty1,phase2:difficulty2) 0.41 1.00
## cor(expected1:difficulty2,phase2:difficulty2) 0.38 1.00
## cor(phase1:difficulty1,phase2:difficulty2) 0.41 1.00
## cor(phase2:difficulty1,phase2:difficulty2) 0.38 1.00
## cor(phase1:difficulty2,phase2:difficulty2) 0.41 1.00
## cor(Intercept,expected1:phase1:difficulty1) 0.44 1.00
## cor(expected1,expected1:phase1:difficulty1) 0.38 1.00
## cor(phase1,expected1:phase1:difficulty1) 0.40 1.00
## cor(phase2,expected1:phase1:difficulty1) 0.46 1.00
## cor(difficulty1,expected1:phase1:difficulty1) 0.42 1.00
## cor(difficulty2,expected1:phase1:difficulty1) 0.41 1.00
## cor(expected1:phase1,expected1:phase1:difficulty1) 0.41 1.00
## cor(expected1:phase2,expected1:phase1:difficulty1) 0.43 1.00
## cor(expected1:difficulty1,expected1:phase1:difficulty1) 0.44 1.00
## cor(expected1:difficulty2,expected1:phase1:difficulty1) 0.42 1.00
## cor(phase1:difficulty1,expected1:phase1:difficulty1) 0.41 1.00
## cor(phase2:difficulty1,expected1:phase1:difficulty1) 0.39 1.00
## cor(phase1:difficulty2,expected1:phase1:difficulty1) 0.41 1.00
## cor(phase2:difficulty2,expected1:phase1:difficulty1) 0.40 1.00
## cor(Intercept,expected1:phase2:difficulty1) 0.49 1.00
## cor(expected1,expected1:phase2:difficulty1) 0.44 1.00
## cor(phase1,expected1:phase2:difficulty1) 0.44 1.00
## cor(phase2,expected1:phase2:difficulty1) 0.48 1.00
## cor(difficulty1,expected1:phase2:difficulty1) 0.42 1.00
## cor(difficulty2,expected1:phase2:difficulty1) 0.42 1.00
## cor(expected1:phase1,expected1:phase2:difficulty1) 0.42 1.00
## cor(expected1:phase2,expected1:phase2:difficulty1) 0.43 1.00
## cor(expected1:difficulty1,expected1:phase2:difficulty1) 0.42 1.00
## cor(expected1:difficulty2,expected1:phase2:difficulty1) 0.42 1.00
## cor(phase1:difficulty1,expected1:phase2:difficulty1) 0.42 1.00
## cor(phase2:difficulty1,expected1:phase2:difficulty1) 0.39 1.00
## cor(phase1:difficulty2,expected1:phase2:difficulty1) 0.42 1.00
## cor(phase2:difficulty2,expected1:phase2:difficulty1) 0.40 1.00
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty1) 0.41 1.00
## cor(Intercept,expected1:phase1:difficulty2) 0.43 1.00
## cor(expected1,expected1:phase1:difficulty2) 0.36 1.00
## cor(phase1,expected1:phase1:difficulty2) 0.36 1.00
## cor(phase2,expected1:phase1:difficulty2) 0.50 1.00
## cor(difficulty1,expected1:phase1:difficulty2) 0.41 1.00
## cor(difficulty2,expected1:phase1:difficulty2) 0.45 1.00
## cor(expected1:phase1,expected1:phase1:difficulty2) 0.38 1.00
## cor(expected1:phase2,expected1:phase1:difficulty2) 0.45 1.00
## cor(expected1:difficulty1,expected1:phase1:difficulty2) 0.41 1.00
## cor(expected1:difficulty2,expected1:phase1:difficulty2) 0.45 1.00
## cor(phase1:difficulty1,expected1:phase1:difficulty2) 0.41 1.00
## cor(phase2:difficulty1,expected1:phase1:difficulty2) 0.41 1.00
## cor(phase1:difficulty2,expected1:phase1:difficulty2) 0.40 1.00
## cor(phase2:difficulty2,expected1:phase1:difficulty2) 0.41 1.00
## cor(expected1:phase1:difficulty1,expected1:phase1:difficulty2) 0.41 1.00
## cor(expected1:phase2:difficulty1,expected1:phase1:difficulty2) 0.43 1.00
## cor(Intercept,expected1:phase2:difficulty2) 0.49 1.00
## cor(expected1,expected1:phase2:difficulty2) 0.41 1.00
## cor(phase1,expected1:phase2:difficulty2) 0.38 1.00
## cor(phase2,expected1:phase2:difficulty2) 0.49 1.00
## cor(difficulty1,expected1:phase2:difficulty2) 0.42 1.00
## cor(difficulty2,expected1:phase2:difficulty2) 0.44 1.00

```

```

## cor(expected1:phase1,expected1:phase2:difficulty2)          0.43 1.00
## cor(expected1:phase2,expected1:phase2:difficulty2)          0.45 1.00
## cor(expected1:difficulty1,expected1:phase2:difficulty2)      0.40 1.00
## cor(expected1:difficulty2,expected1:phase2:difficulty2)      0.38 1.00
## cor(phase1:difficulty1,expected1:phase2:difficulty2)         0.42 1.00
## cor(phase2:difficulty1,expected1:phase2:difficulty2)         0.39 1.00
## cor(phase1:difficulty2,expected1:phase2:difficulty2)         0.43 1.00
## cor(phase2:difficulty2,expected1:phase2:difficulty2)         0.41 1.00
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty2) 0.43 1.00
## cor(expected1:phase2:difficulty1,expected1:phase2:difficulty2) 0.41 1.00
## cor(expected1:phase1:difficulty2,expected1:phase2:difficulty2) 0.43 1.00
## Bulk_ESS
## sd(Intercept)          3149
## sd(expected1)          4745
## sd(phase1)             7678
## sd(phase2)             7995
## sd(difficulty1)        5760
## sd(difficulty2)        6791
## sd(expected1:phase1)   4821
## sd(expected1:phase2)   5887
## sd(expected1:difficulty1) 9067
## sd(expected1:difficulty2) 6088
## sd(phase1:difficulty1) 10782
## sd(phase2:difficulty1)  4812
## sd(phase1:difficulty2)  8954
## sd(phase2:difficulty2)  5878
## sd(expected1:phase1:difficulty1) 8722
## sd(expected1:phase2:difficulty1) 8523
## sd(expected1:phase1:difficulty2) 6855
## sd(expected1:phase2:difficulty2) 8083
## cor(Intercept,expected1) 27289
## cor(Intercept,phase1)   9349
## cor(expected1,phase1)   2613
## cor(Intercept,phase2) 18825
## cor(expected1,phase2)   8621
## cor(phase1,phase2)      15074
## cor(Intercept,difficulty1) 34860
## cor(expected1,difficulty1) 17919
## cor(phase1,difficulty1)  21107
## cor(phase2,difficulty1)  25935
## cor(Intercept,difficulty2) 35309
## cor(expected1,difficulty2) 24094
## cor(phase1,difficulty2)  28374
## cor(phase2,difficulty2)  27923
## cor(difficulty1,difficulty2) 20285
## cor(Intercept,expected1:phase1) 35442
## cor(expected1,expected1:phase1) 19747
## cor(phase1,expected1:phase1)  30130
## cor(phase2,expected1:phase1)  20802
## cor(difficulty1,expected1:phase1) 17965
## cor(difficulty2,expected1:phase1) 16948
## cor(Intercept,expected1:phase2) 34728
## cor(expected1,expected1:phase2) 26274
## cor(phase1,expected1:phase2)  27911
## cor(phase2,expected1:phase2)  22263
## cor(difficulty1,expected1:phase2) 19179
## cor(difficulty2,expected1:phase2) 14220
## cor(expected1:phase1,expected1:phase2) 17477
## cor(Intercept,expected1:difficulty1) 35774
## cor(expected1,expected1:difficulty1) 31034

```

## cor(phase1,expected1:difficulty1)	31936
## cor(phase2,expected1:difficulty1)	27614
## cor(difficulty1,expected1:difficulty1)	21439
## cor(difficulty2,expected1:difficulty1)	20297
## cor(expected1:phase1,expected1:difficulty1)	19323
## cor(expected1:phase2,expected1:difficulty1)	15576
## cor(Intercept,expected1:difficulty2)	26699
## cor(expected1,expected1:difficulty2)	12358
## cor(phase1,expected1:difficulty2)	18573
## cor(phase2,expected1:difficulty2)	19170
## cor(difficulty1,expected1:difficulty2)	12139
## cor(difficulty2,expected1:difficulty2)	14601
## cor(expected1:phase1,expected1:difficulty2)	13726
## cor(expected1:phase2,expected1:difficulty2)	11892
## cor(expected1:difficulty1,expected1:difficulty2)	12103
## cor(Intercept,phase1:difficulty1)	37352
## cor(expected1,phase1:difficulty1)	32823
## cor(phase1,phase1:difficulty1)	35470
## cor(phase2,phase1:difficulty1)	27977
## cor(difficulty1,phase1:difficulty1)	21646
## cor(difficulty2,phase1:difficulty1)	19556
## cor(expected1:phase1,phase1:difficulty1)	18786
## cor(expected1:phase2,phase1:difficulty1)	17036
## cor(expected1:difficulty1,phase1:difficulty1)	14101
## cor(expected1:difficulty2,phase1:difficulty1)	18993
## cor(Intercept,phase2:difficulty1)	28050
## cor(expected1,phase2:difficulty1)	14117
## cor(phase1,phase2:difficulty1)	25198
## cor(phase2,phase2:difficulty1)	20726
## cor(difficulty1,phase2:difficulty1)	13103
## cor(difficulty2,phase2:difficulty1)	13036
## cor(expected1:phase1,phase2:difficulty1)	13805
## cor(expected1:phase2,phase2:difficulty1)	14182
## cor(expected1:difficulty1,phase2:difficulty1)	13134
## cor(expected1:difficulty2,phase2:difficulty1)	15978
## cor(phase1:difficulty1,phase2:difficulty1)	12604
## cor(Intercept,phase1:difficulty2)	34092
## cor(expected1,phase1:difficulty2)	30039
## cor(phase1,phase1:difficulty2)	31498
## cor(phase2,phase1:difficulty2)	25545
## cor(difficulty1,phase1:difficulty2)	20397
## cor(difficulty2,phase1:difficulty2)	19287
## cor(expected1:phase1,phase1:difficulty2)	18012
## cor(expected1:phase2,phase1:difficulty2)	16964
## cor(expected1:difficulty1,phase1:difficulty2)	13838
## cor(expected1:difficulty2,phase1:difficulty2)	18331
## cor(phase1:difficulty1,phase1:difficulty2)	13161
## cor(phase2:difficulty1,phase1:difficulty2)	15761
## cor(Intercept,phase2:difficulty2)	31526
## cor(expected1,phase2:difficulty2)	24938
## cor(phase1,phase2:difficulty2)	23884
## cor(phase2,phase2:difficulty2)	25363
## cor(difficulty1,phase2:difficulty2)	18090
## cor(difficulty2,phase2:difficulty2)	16719
## cor(expected1:phase1,phase2:difficulty2)	17322
## cor(expected1:phase2,phase2:difficulty2)	15872
## cor(expected1:difficulty1,phase2:difficulty2)	14167
## cor(expected1:difficulty2,phase2:difficulty2)	18084
## cor(phase1:difficulty1,phase2:difficulty2)	12081
## cor(phase2:difficulty1,phase2:difficulty2)	14874

## cor(phase1:difficulty2,phase2:difficulty2)	12833
## cor(Intercept,expected1:phase1:difficulty1)	33605
## cor(expected1,expected1:phase1:difficulty1)	27705
## cor(phase1,expected1:phase1:difficulty1)	35093
## cor(phase2,expected1:phase1:difficulty1)	27745
## cor(difficulty1,expected1:phase1:difficulty1)	21512
## cor(difficulty2,expected1:phase1:difficulty1)	20067
## cor(expected1:phase1,expected1:phase1:difficulty1)	18415
## cor(expected1:phase2,expected1:phase1:difficulty1)	16294
## cor(expected1:difficulty1,expected1:phase1:difficulty1)	15096
## cor(expected1:difficulty2,expected1:phase1:difficulty1)	20155
## cor(phase1:difficulty1,expected1:phase1:difficulty1)	12484
## cor(phase2:difficulty1,expected1:phase1:difficulty1)	15997
## cor(phase1:difficulty2,expected1:phase1:difficulty1)	11619
## cor(phase2:difficulty2,expected1:phase1:difficulty1)	12280
## cor(Intercept,expected1:phase2:difficulty1)	26990
## cor(expected1,expected1:phase2:difficulty1)	27447
## cor(phase1,expected1:phase2:difficulty1)	30311
## cor(phase2,expected1:phase2:difficulty1)	22611
## cor(difficulty1,expected1:phase2:difficulty1)	20483
## cor(difficulty2,expected1:phase2:difficulty1)	19350
## cor(expected1:phase1,expected1:phase2:difficulty1)	17896
## cor(expected1:phase2,expected1:phase2:difficulty1)	17484
## cor(expected1:difficulty1,expected1:phase2:difficulty1)	14938
## cor(expected1:difficulty2,expected1:phase2:difficulty1)	20080
## cor(phase1:difficulty1,expected1:phase2:difficulty1)	12491
## cor(phase2:difficulty1,expected1:phase2:difficulty1)	15188
## cor(phase1:difficulty2,expected1:phase2:difficulty1)	11531
## cor(phase2:difficulty2,expected1:phase2:difficulty1)	12837
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty1)	10815
## cor(Intercept,expected1:phase1:difficulty2)	34730
## cor(expected1,expected1:phase1:difficulty2)	23642
## cor(phase1,expected1:phase1:difficulty2)	28945
## cor(phase2,expected1:phase1:difficulty2)	20249
## cor(difficulty1,expected1:phase1:difficulty2)	19707
## cor(difficulty2,expected1:phase1:difficulty2)	18108
## cor(expected1:phase1,expected1:phase1:difficulty2)	15806
## cor(expected1:phase2,expected1:phase1:difficulty2)	14364
## cor(expected1:difficulty1,expected1:phase1:difficulty2)	15408
## cor(expected1:difficulty2,expected1:phase1:difficulty2)	18049
## cor(phase1:difficulty1,expected1:phase1:difficulty2)	13491
## cor(phase2:difficulty1,expected1:phase1:difficulty2)	17107
## cor(phase1:difficulty2,expected1:phase1:difficulty2)	12272
## cor(phase2:difficulty2,expected1:phase1:difficulty2)	12874
## cor(expected1:phase1:difficulty1,expected1:phase1:difficulty2)	11083
## cor(expected1:phase2:difficulty1,expected1:phase1:difficulty2)	11112
## cor(Intercept,expected1:phase2:difficulty2)	28133
## cor(expected1,expected1:phase2:difficulty2)	28291
## cor(phase1,expected1:phase2:difficulty2)	31934
## cor(phase2,expected1:phase2:difficulty2)	25747
## cor(difficulty1,expected1:phase2:difficulty2)	21361
## cor(difficulty2,expected1:phase2:difficulty2)	18911
## cor(expected1:phase1,expected1:phase2:difficulty2)	19111
## cor(expected1:phase2,expected1:phase2:difficulty2)	16030
## cor(expected1:difficulty1,expected1:phase2:difficulty2)	13591
## cor(expected1:difficulty2,expected1:phase2:difficulty2)	19047
## cor(phase1:difficulty1,expected1:phase2:difficulty2)	12615
## cor(phase2:difficulty1,expected1:phase2:difficulty2)	14862
## cor(phase1:difficulty2,expected1:phase2:difficulty2)	12112
## cor(phase2:difficulty2,expected1:phase2:difficulty2)	13632

## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty2)	11191
## cor(expected1:phase2:difficulty1,expected1:phase2:difficulty2)	9659
## cor(expected1:phase1:difficulty2,expected1:phase2:difficulty2)	11249
##	Tail_ESS
## sd(Intercept)	5544
## sd(expected1)	4084
## sd(phase1)	11763
## sd(phase2)	9176
## sd(difficulty1)	7605
## sd(difficulty2)	8670
## sd(expected1:phase1)	8183
## sd(expected1:phase2)	8133
## sd(expected1:difficulty1)	8851
## sd(expected1:difficulty2)	4343
## sd(phase1:difficulty1)	9577
## sd(phase2:difficulty1)	5235
## sd(phase1:difficulty2)	8480
## sd(phase2:difficulty2)	7346
## sd(expected1:phase1:difficulty1)	8137
## sd(expected1:phase2:difficulty1)	9658
## sd(expected1:phase1:difficulty2)	7577
## sd(expected1:phase2:difficulty2)	8810
## cor(Intercept,expected1)	14201
## cor(Intercept,phase1)	11917
## cor(expected1,phase1)	5125
## cor(Intercept,phase2)	14694
## cor(expected1,phase2)	11478
## cor(phase1,phase2)	14457
## cor(Intercept,difficulty1)	14221
## cor(expected1,difficulty1)	12664
## cor(phase1,difficulty1)	14083
## cor(phase2,difficulty1)	13281
## cor(Intercept,difficulty2)	12870
## cor(expected1,difficulty2)	14092
## cor(phase1,difficulty2)	14199
## cor(phase2,difficulty2)	14664
## cor(difficulty1,difficulty2)	13949
## cor(Intercept,expected1:phase1)	12829
## cor(expected1,expected1:phase1)	13463
## cor(phase1,expected1:phase1)	13750
## cor(phase2,expected1:phase1)	13930
## cor(difficulty1,expected1:phase1)	14732
## cor(difficulty2,expected1:phase1)	15016
## cor(Intercept,expected1:phase2)	13878
## cor(expected1,expected1:phase2)	13258
## cor(phase1,expected1:phase2)	13628
## cor(phase2,expected1:phase2)	14212
## cor(difficulty1,expected1:phase2)	14839
## cor(difficulty2,expected1:phase2)	13554
## cor(expected1:phase1,expected1:phase2)	15153
## cor(Intercept,expected1:difficulty1)	13338
## cor(expected1,expected1:difficulty1)	13438
## cor(phase1,expected1:difficulty1)	12880
## cor(phase2,expected1:difficulty1)	13220
## cor(difficulty1,expected1:difficulty1)	14011
## cor(difficulty2,expected1:difficulty1)	14546
## cor(expected1:phase1,expected1:difficulty1)	14257
## cor(expected1:phase2,expected1:difficulty1)	12705
## cor(Intercept,expected1:difficulty2)	13952
## cor(expected1,expected1:difficulty2)	13583

## cor(phase1,expected1:difficulty2)	12697
## cor(phase2,expected1:difficulty2)	14952
## cor(difficulty1,expected1:difficulty2)	14318
## cor(difficulty2,expected1:difficulty2)	14570
## cor(expected1:phase1,expected1:difficulty2)	15096
## cor(expected1:phase2,expected1:difficulty2)	14788
## cor(expected1:difficulty1,expected1:difficulty2)	15125
## cor(Intercept,phase1:difficulty1)	13678
## cor(expected1,phase1:difficulty1)	14233
## cor(phase1,phase1:difficulty1)	13191
## cor(phase2,phase1:difficulty1)	13129
## cor(difficulty1,phase1:difficulty1)	13692
## cor(difficulty2,phase1:difficulty1)	14200
## cor(expected1:phase1,phase1:difficulty1)	14026
## cor(expected1:phase2,phase1:difficulty1)	14219
## cor(expected1:difficulty1,phase1:difficulty1)	14254
## cor(expected1:difficulty2,phase1:difficulty1)	15218
## cor(Intercept,phase2:difficulty1)	12762
## cor(expected1,phase2:difficulty1)	13558
## cor(phase1,phase2:difficulty1)	14444
## cor(phase2,phase2:difficulty1)	13904
## cor(difficulty1,phase2:difficulty1)	13969
## cor(difficulty2,phase2:difficulty1)	14586
## cor(expected1:phase1,phase2:difficulty1)	14625
## cor(expected1:phase2,phase2:difficulty1)	15007
## cor(expected1:difficulty1,phase2:difficulty1)	13761
## cor(expected1:difficulty2,phase2:difficulty1)	16048
## cor(phase1:difficulty1,phase2:difficulty1)	14648
## cor(Intercept,phase1:difficulty2)	13578
## cor(expected1,phase1:difficulty2)	12528
## cor(phase1,phase1:difficulty2)	13553
## cor(phase2,phase1:difficulty2)	12976
## cor(difficulty1,phase1:difficulty2)	13062
## cor(difficulty2,phase1:difficulty2)	13990
## cor(expected1:phase1,phase1:difficulty2)	14922
## cor(expected1:phase2,phase1:difficulty2)	15245
## cor(expected1:difficulty1,phase1:difficulty2)	13665
## cor(expected1:difficulty2,phase1:difficulty2)	14660
## cor(phase1:difficulty1,phase1:difficulty2)	14547
## cor(phase2:difficulty1,phase1:difficulty2)	15387
## cor(Intercept,phase2:difficulty2)	13859
## cor(expected1,phase2:difficulty2)	14301
## cor(phase1,phase2:difficulty2)	12800
## cor(phase2,phase2:difficulty2)	14350
## cor(difficulty1,phase2:difficulty2)	14777
## cor(difficulty2,phase2:difficulty2)	14355
## cor(expected1:phase1,phase2:difficulty2)	14374
## cor(expected1:phase2,phase2:difficulty2)	15093
## cor(expected1:difficulty1,phase2:difficulty2)	15077
## cor(expected1:difficulty2,phase2:difficulty2)	15242
## cor(phase1:difficulty1,phase2:difficulty2)	14853
## cor(phase2:difficulty1,phase2:difficulty2)	15856
## cor(phase1:difficulty2,phase2:difficulty2)	14907
## cor(Intercept,expected1:phase1:difficulty1)	13388
## cor(expected1,expected1:phase1:difficulty1)	14077
## cor(phase1,expected1:phase1:difficulty1)	13484
## cor(phase2,expected1:phase1:difficulty1)	13443
## cor(difficulty1,expected1:phase1:difficulty1)	14630
## cor(difficulty2,expected1:phase1:difficulty1)	13979
## cor(expected1:phase1,expected1:phase1:difficulty1)	15474

```

## cor(expected1:phase2,expected1:phase1:difficulty1) 14892
## cor(expected1:difficulty1,expected1:phase1:difficulty1) 15168
## cor(expected1:difficulty2,expected1:phase1:difficulty1) 14844
## cor(phase1:difficulty1,expected1:phase1:difficulty1) 14111
## cor(phase2:difficulty1,expected1:phase1:difficulty1) 15451
## cor(phase1:difficulty2,expected1:phase1:difficulty1) 13985
## cor(phase2:difficulty2,expected1:phase1:difficulty1) 14042
## cor(Intercept,expected1:phase2:difficulty1) 13634
## cor(expected1,expected1:phase2:difficulty1) 13657
## cor(phase1,expected1:phase2:difficulty1) 13391
## cor(phase2,expected1:phase2:difficulty1) 13987
## cor(difficulty1,expected1:phase2:difficulty1) 13551
## cor(difficulty2,expected1:phase2:difficulty1) 14253
## cor(expected1:phase1,expected1:phase2:difficulty1) 14080
## cor(expected1:phase2,expected1:phase2:difficulty1) 14275
## cor(expected1:difficulty1,expected1:phase2:difficulty1) 14949
## cor(expected1:difficulty2,expected1:phase2:difficulty1) 15395
## cor(phase1:difficulty1,expected1:phase2:difficulty1) 14651
## cor(phase2:difficulty1,expected1:phase2:difficulty1) 14521
## cor(phase1:difficulty2,expected1:phase2:difficulty1) 13935
## cor(phase2:difficulty2,expected1:phase2:difficulty1) 14099
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty1) 14314
## cor(Intercept,expected1:phase1:difficulty2) 12487
## cor(expected1,expected1:phase1:difficulty2) 14044
## cor(phase1,expected1:phase1:difficulty2) 13872
## cor(phase2,expected1:phase1:difficulty2) 13094
## cor(difficulty1,expected1:phase1:difficulty2) 13371
## cor(difficulty2,expected1:phase1:difficulty2) 14929
## cor(expected1:phase1,expected1:phase1:difficulty2) 15211
## cor(expected1:phase2,expected1:phase1:difficulty2) 14118
## cor(expected1:difficulty1,expected1:phase1:difficulty2) 15059
## cor(expected1:difficulty2,expected1:phase1:difficulty2) 15279
## cor(phase1:difficulty1,expected1:phase1:difficulty2) 15251
## cor(phase2:difficulty1,expected1:phase1:difficulty2) 15467
## cor(phase1:difficulty2,expected1:phase1:difficulty2) 14909
## cor(phase2:difficulty2,expected1:phase1:difficulty2) 14660
## cor(expected1:phase1:difficulty1,expected1:phase1:difficulty2) 14105
## cor(expected1:phase2:difficulty1,expected1:phase1:difficulty2) 14394
## cor(Intercept,expected1:phase2:difficulty2) 13149
## cor(expected1,expected1:phase2:difficulty2) 13706
## cor(phase1,expected1:phase2:difficulty2) 13551
## cor(phase2,expected1:phase2:difficulty2) 13922
## cor(difficulty1,expected1:phase2:difficulty2) 14221
## cor(difficulty2,expected1:phase2:difficulty2) 14651
## cor(expected1:phase1,expected1:phase2:difficulty2) 14390
## cor(expected1:phase2,expected1:phase2:difficulty2) 14380
## cor(expected1:difficulty1,expected1:phase2:difficulty2) 14311
## cor(expected1:difficulty2,expected1:phase2:difficulty2) 15488
## cor(phase1:difficulty1,expected1:phase2:difficulty2) 14330
## cor(phase2:difficulty1,expected1:phase2:difficulty2) 13690
## cor(phase1:difficulty2,expected1:phase2:difficulty2) 13794
## cor(phase2:difficulty2,expected1:phase2:difficulty2) 15011
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty2) 15334
## cor(expected1:phase2:difficulty1,expected1:phase2:difficulty2) 13620
## cor(expected1:phase1:difficulty2,expected1:phase2:difficulty2) 13966
##
## ~trl (Number of levels: 288)
##
## Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS
## sd(Intercept) 0.06 0.00 0.05 0.07 1.00 7395
## sd(diagnosis1) 0.01 0.00 0.00 0.02 1.00 3741

```

```

## cor(Intercept,diagnosis1)      0.03      0.35      -0.69      0.71 1.00      21237
##                               Tail_ESS
## sd(Intercept)                  12182
## sd(diagnosis1)                  7318
## cor(Intercept,diagnosis1)      10591
##
## Regression Coefficients:
##                               Estimate Est.Error 1-95% CI u-95% CI
## Intercept                      6.17      0.03      6.12      6.23
## diagnosis1                     0.03      0.02     -0.01      0.07
## expected1                     -0.04      0.01     -0.05     -0.02
## phase1                         0.03      0.01      0.00      0.05
## phase2                       -0.01      0.01     -0.02      0.01
## difficulty1                   -0.03      0.01     -0.04     -0.01
## difficulty2                   -0.01      0.01     -0.02      0.01
## diagnosis1:expected1          -0.00      0.00     -0.01      0.01
## diagnosis1:phase1             0.00      0.01     -0.02      0.02
## diagnosis1:phase2            -0.00      0.01     -0.01      0.01
## expected1:phase1             -0.04      0.01     -0.05     -0.02
## expected1:phase2             0.00      0.01     -0.01      0.02
## diagnosis1:difficulty1        0.00      0.00     -0.01      0.01
## diagnosis1:difficulty2       -0.00      0.00     -0.01      0.01
## expected1:difficulty1        -0.01      0.01     -0.03      0.00
## expected1:difficulty2        0.00      0.01     -0.01      0.02
## phase1:difficulty1           -0.00      0.01     -0.02      0.02
## phase2:difficulty1           0.01      0.01     -0.01      0.03
## phase1:difficulty2           0.02      0.01     -0.00      0.04
## phase2:difficulty2          -0.01      0.01     -0.03      0.01
## diagnosis1:expected1:phase1   0.01      0.01     -0.00      0.02
## diagnosis1:expected1:phase2   0.00      0.00     -0.01      0.01
## diagnosis1:expected1:difficulty1 -0.00      0.00     -0.01      0.01
## diagnosis1:expected1:difficulty2 0.00      0.00     -0.01      0.01
## diagnosis1:phase1:difficulty1 0.01      0.01     -0.01      0.02
## diagnosis1:phase2:difficulty1 -0.00      0.01     -0.01      0.01
## diagnosis1:phase1:difficulty2 -0.01      0.01     -0.02      0.01
## diagnosis1:phase2:difficulty2 0.01      0.01     -0.01      0.02
## expected1:phase1:difficulty1 -0.01      0.01     -0.03      0.01
## expected1:phase2:difficulty1 -0.00      0.01     -0.02      0.02
## expected1:phase1:difficulty2 0.01      0.01     -0.02      0.03
## expected1:phase2:difficulty2 -0.00      0.01     -0.02      0.02
## diagnosis1:expected1:phase1:difficulty1 0.00      0.01     -0.01      0.02
## diagnosis1:expected1:phase2:difficulty1 0.00      0.01     -0.01      0.01
## diagnosis1:expected1:phase1:difficulty2 -0.00      0.01     -0.01      0.01
## diagnosis1:expected1:phase2:difficulty2 -0.00      0.01     -0.02      0.01
##
##                               Rhat Bulk_ESS Tail_ESS
## Intercept                      1.00      1845      4403
## diagnosis1                     1.00      1884      3305
## expected1                     1.00     10083     12702
## phase1                        1.00      6761      9589
## phase2                        1.00      8441     11148
## difficulty1                   1.00      8289     11626
## difficulty2                   1.00      8504     11731
## diagnosis1:expected1          1.00     19211     14759
## diagnosis1:phase1             1.00      6434     10799
## diagnosis1:phase2             1.00     15067     14633
## expected1:phase1              1.00      9181     11343
## expected1:phase2              1.00      8635     12070
## diagnosis1:difficulty1        1.00     16956     14624
## diagnosis1:difficulty2        1.00     18692     15227
## expected1:difficulty1         1.00      8002     11088

```

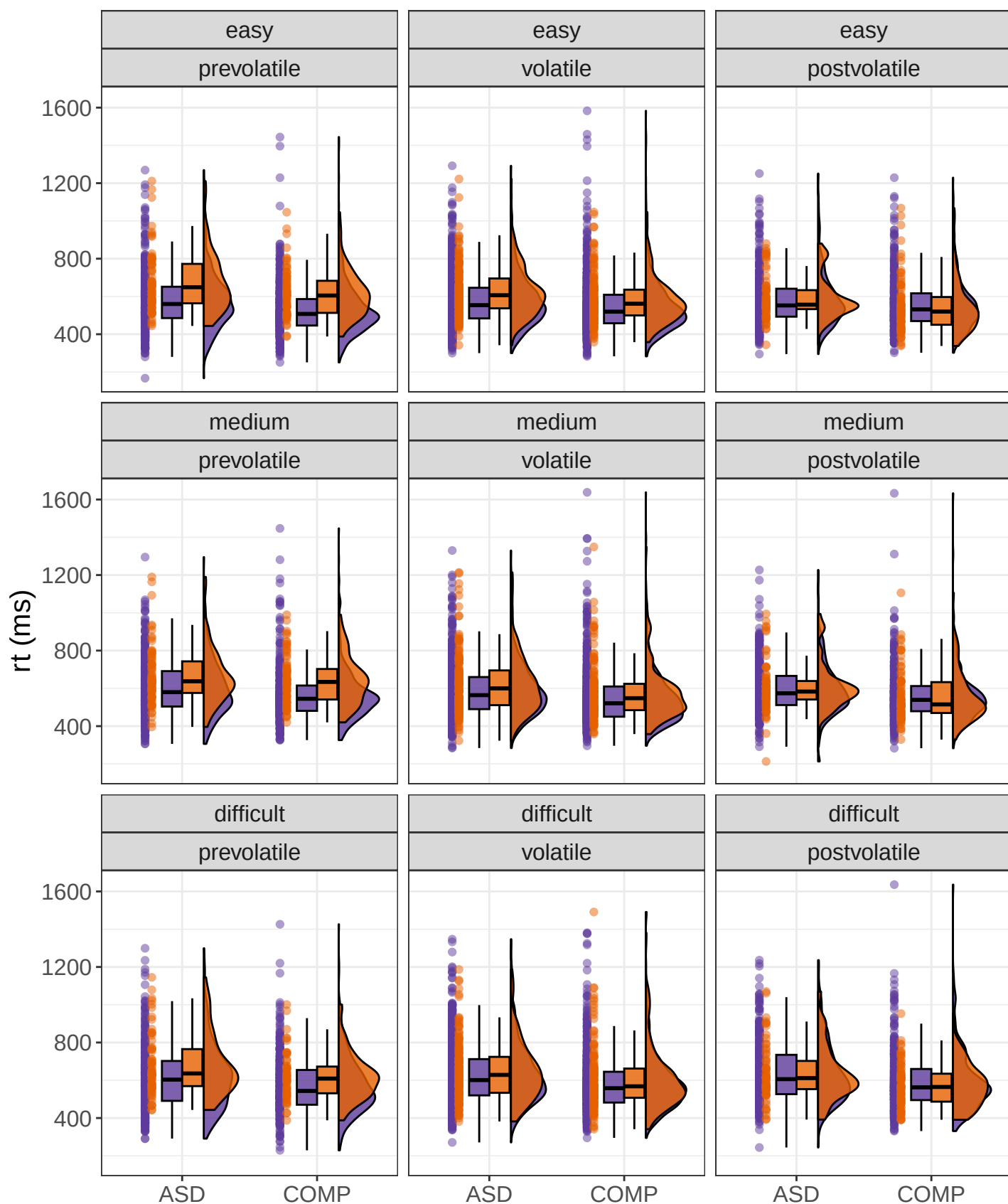
```

## expected1:difficulty2          1.00    7965    10283
## phase1:difficulty1             1.00    8060    11648
## phase2:difficulty1             1.00    7989    11209
## phase1:difficulty2             1.00    8367    11425
## phase2:difficulty2             1.00    7970    11218
## diagnosis1:expected1:phase1    1.00   25920   15621
## diagnosis1:expected1:phase2    1.00   23007   15510
## diagnosis1:expected1:difficulty1 1.00   17794   14751
## diagnosis1:expected1:difficulty2 1.00   16122   14948
## diagnosis1:phase1:difficulty1  1.00   15240   14451
## diagnosis1:phase2:difficulty1  1.00   13895   12915
## diagnosis1:phase1:difficulty2  1.00   15980   14790
## diagnosis1:phase2:difficulty2  1.00   15110   14708
## expected1:phase1:difficulty1    1.00    7790   11261
## expected1:phase2:difficulty1    1.00    7602   11295
## expected1:phase1:difficulty2    1.00    8608   12268
## expected1:phase2:difficulty2    1.00    8034   10958
## diagnosis1:expected1:phase1:difficulty1 1.00   15527   13584
## diagnosis1:expected1:phase2:difficulty1 1.00   14325   14873
## diagnosis1:expected1:phase1:difficulty2 1.00   16115   14760
## diagnosis1:expected1:phase2:difficulty2 1.00   15825   14645
##
## Further Distributional Parameters:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma      0.23      0.00   0.22   0.24 1.00   13158   13192
## ndt       100.15      7.50  84.81 114.20 1.00   12727   13392
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

## Warning in (function (mapping = NULL, data = NULL, stat = "half_ydensity", :
## Ignoring unknown parameters: `outlier.shape`

```

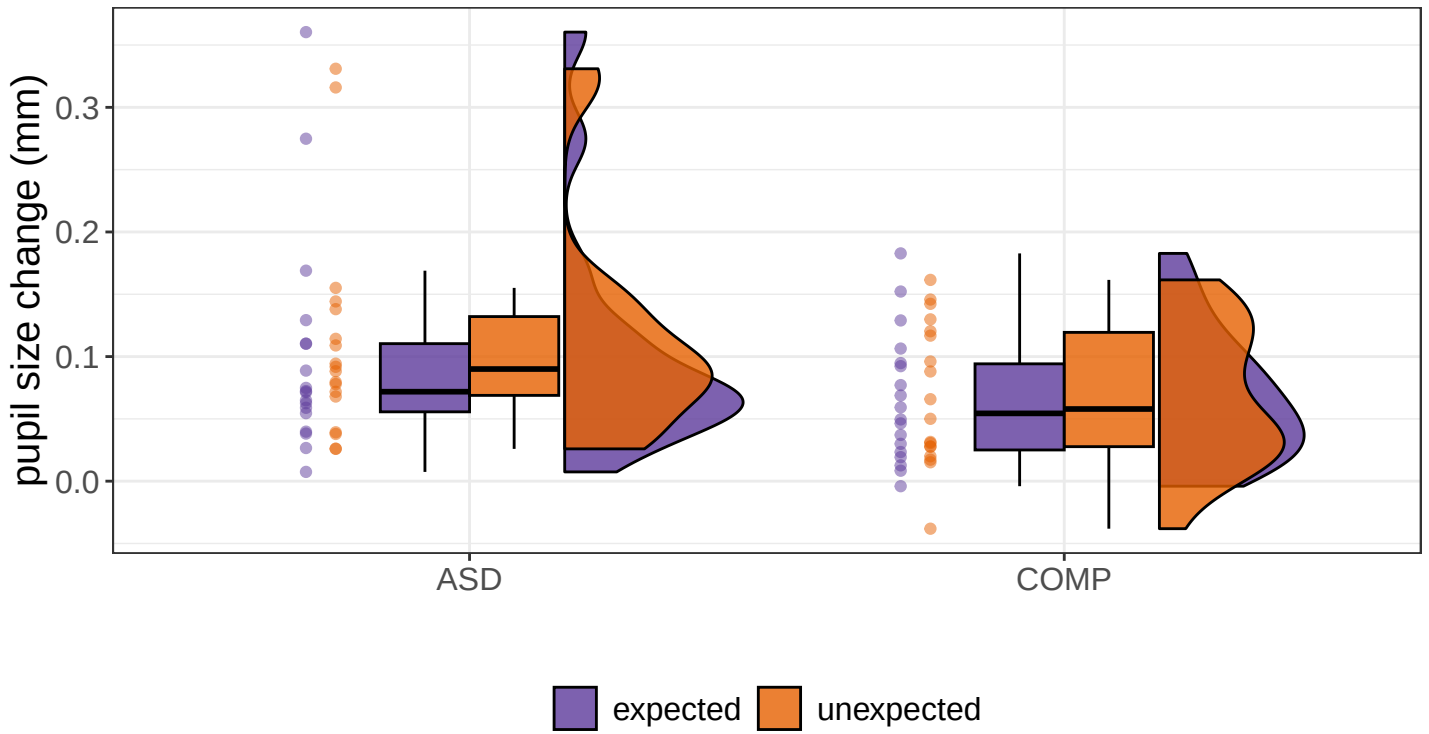
Reaction times per subject



expected unexpected

S5.2 Pupil sizes

```
## Family: gaussian
## Links: mu = identity
## Formula: rel_pupil ~ diagnosis * expected + rts + (1 | subID)
## Data: df (Number of observations: 9386)
## Draws: 4 chains, each with iter = 6000; warmup = 1500; thin = 1;
## total post-warmup draws = 18000
##
## Multilevel Hyperparameters:
## ~subID (Number of levels: 36)
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)      0.08      0.01      0.06      0.10 1.00      1241      2664
##
## Regression Coefficients:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS
## Intercept          0.09      0.01      0.06      0.11 1.01        665
## diagnosis1          0.01      0.01     -0.01      0.04 1.01        519
## expected1         -0.00      0.00     -0.01      0.00 1.00      14324
## rts                 0.00      0.00      0.00      0.00 1.00      19381
## diagnosis1:expected1 -0.00      0.00     -0.01      0.00 1.00      14336
##      Tail_ESS
## Intercept          1481
## diagnosis1          1096
## expected1          13389
## rts                 14510
## diagnosis1:expected1 13538
##
## Further Distributional Parameters:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma      0.16      0.00      0.16      0.16 1.00      7253      9039
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
##
## Warning in (function (mapping = NULL, data = NULL, stat = "half_ydensity", :
## Ignoring unknown parameters: `outlier.shape`
```



S5.3 Accuracies

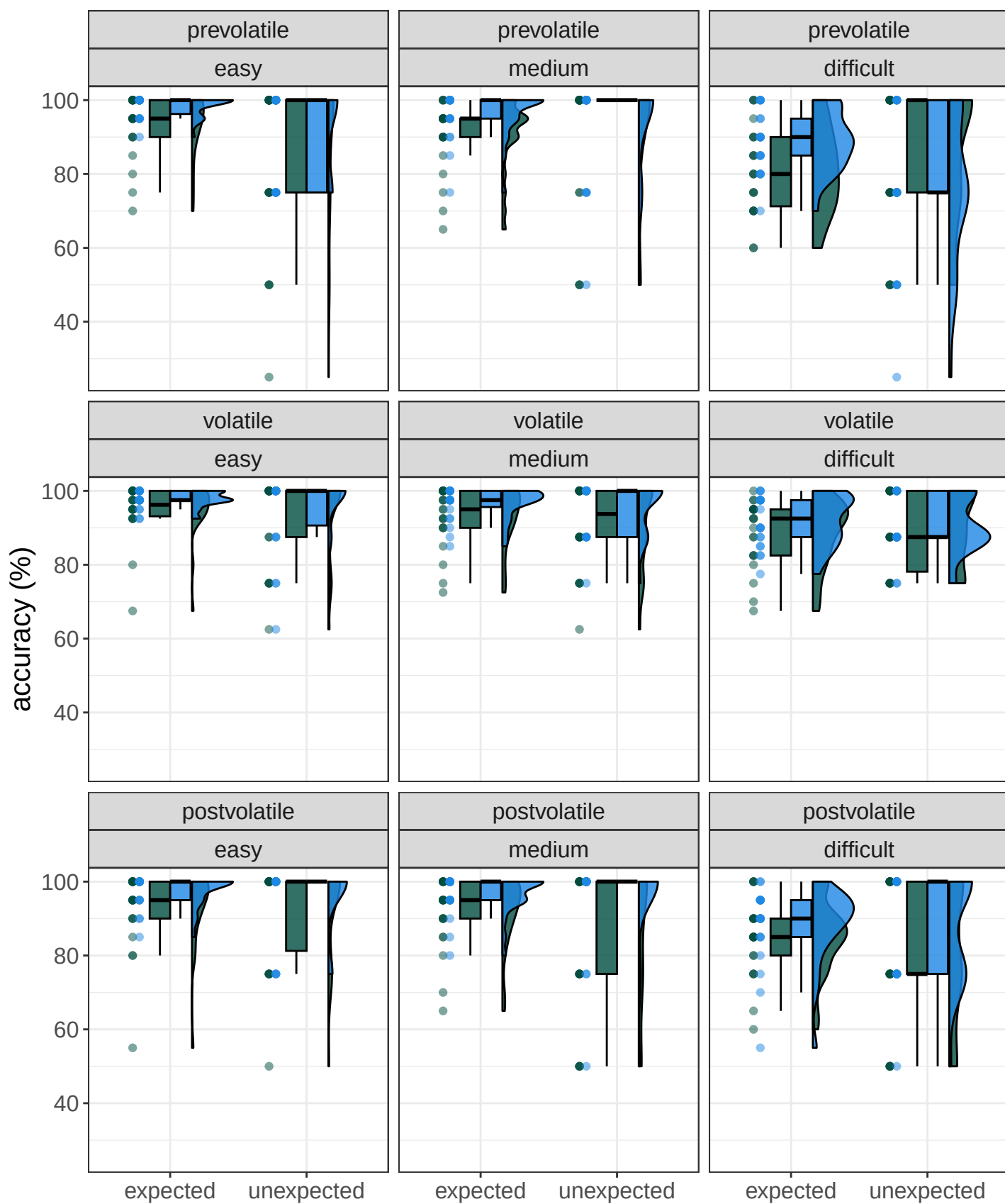
```
## ---
## Model:
## Type: BFlinearModel, JZS
## Intercept only
## Data types:
## diagnosis : fixed
```

	bf
diagnosis + expected + difficulty	41.15
diagnosis + difficulty	40.83
diagnosis + expected + diagnosis:expected + difficulty	40.29
diagnosis + expected + difficulty + diagnosis:difficulty	39.74
diagnosis + difficulty + diagnosis:difficulty	39.39
diagnosis + expected + diagnosis:expected + difficulty + diagnosis:difficulty	38.93
diagnosis + expected + difficulty + expected:difficulty	38.50
diagnosis + expected + diagnosis:expected + difficulty + expected:difficulty	37.72
diagnosis + phase + expected + difficulty	37.37
diagnosis + expected + difficulty + diagnosis:difficulty + expected:difficulty	37.13
diagnosis + phase + difficulty	37.12
diagnosis + phase + expected + diagnosis:expected + difficulty	36.56
diagnosis + expected + diagnosis:expected + difficulty + diagnosis:difficulty + expected:difficulty	36.26
diagnosis + phase + expected + difficulty + diagnosis:difficulty	36.10
diagnosis + phase + difficulty + diagnosis:difficulty	35.64
diagnosis + phase + expected + diagnosis:expected + difficulty + diagnosis:difficulty	35.11
diagnosis + phase + expected + difficulty + expected:difficulty	34.79
diagnosis + phase + diagnosis:phase + expected + difficulty	34.55
diagnosis + phase + diagnosis:phase + difficulty	34.25
diagnosis + phase + expected + phase:expected + difficulty	33.91
diagnosis + phase + expected + diagnosis:expected + difficulty + expected:difficulty	33.87
expected + difficulty	33.81
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + difficulty	33.67
diagnosis + expected + diagnosis:expected + difficulty + diagnosis:difficulty + expected:difficulty + diagnosis:expected:difficulty	33.63

	bf
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty + diagnosis:expected:difficulty + phase:expected:difficulty	16.73
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty + diagnosis:expected:difficulty + phase:expected:difficulty	16.68
diagnosis + phase + diagnosis:phase + expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + phase:expected:difficulty	16.62
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + diagnosis:expected:difficulty	16.25
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + diagnosis:expected:difficulty	16.04
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + diagnosis:expected:difficulty	15.62
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + diagnosis:expected:difficulty	15.46
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + diagnosis:expected:difficulty	13.50
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + diagnosis:expected:difficulty	13.06
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + diagnosis:expected:difficulty	11.21
diagnosis + expected	6.48
diagnosis	6.40
diagnosis + expected + diagnosis:expected	5.52
diagnosis + phase + expected	2.63
diagnosis + phase	2.58
diagnosis + phase + expected + diagnosis:expected	1.74
expected	0.01
diagnosis + phase + diagnosis:phase + expected	-
diagnosis + phase + diagnosis:phase	0.27
diagnosis + phase + expected + phase:expected	-
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected	0.38
diagnosis + phase + expected + diagnosis:expected + phase:expected	-
diagnosis + phase + expected + diagnosis:expected + phase:expected	0.87
diagnosis + phase + diagnosis:phase + expected + phase:expected	-
diagnosis + phase + diagnosis:phase + expected + phase:expected	1.25
diagnosis + phase + diagnosis:phase + expected + phase:expected	-
diagnosis + phase + diagnosis:phase + expected + phase:expected	1.89
diagnosis + phase + diagnosis:phase + expected + phase:expected	-
diagnosis + phase + diagnosis:phase + expected + phase:expected	3.78
phase	-
phase + expected	3.82
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected	-
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected	3.85
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected	-
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected	4.80
phase + expected + phase:expected	-
phase + expected + phase:expected	5.23
phase + expected + phase:expected	-
phase + expected + phase:expected	7.32

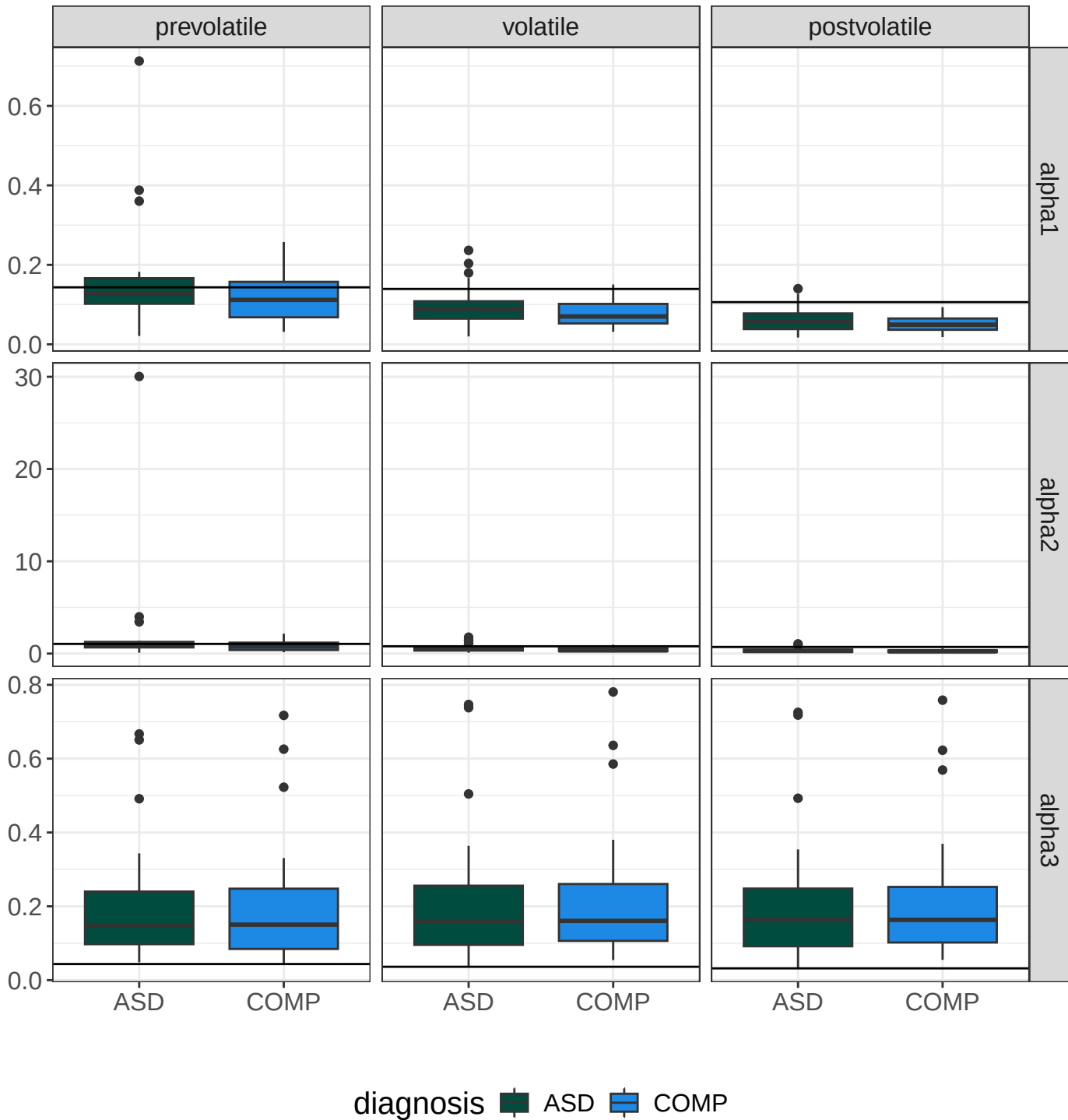
```
## Warning in (function (mapping = NULL, data = NULL, stat = "half_ydensity", :
## Ignoring unknown parameters: `outlier.shape`
```

Accuracies per subject



S6 Learning rates proper

Learning rates



S7 Handedness and button assignment

We counterbalanced whether the left or the right button was to be clicked to indicate a positive valence. Here, we check whether there were any differences between groups in terms of handedness and button assignment. Information on handedness of one participant is missing, because they skipped this question.

```
##                bf error                time                code
## Non-indep. (a=1) -2.444793      0 Sun Feb 15 08:47:16 2026 118e771a16f2
```

diagnosis	handedness	n
ASD	beidh ndig	1
ASD	links	3
ASD	rechts	17
ASD	NA	1
COMP	links	3
COMP	rechts	19

diagnosis	PAL.pos	handedness	n
ASD	left	beidh ndig	1
ASD	left	links	1
ASD	left	rechts	9
ASD	right	links	2
ASD	right	rechts	8
ASD	right	NA	1
COMP	left	links	2
COMP	left	rechts	9
COMP	right	links	1
COMP	right	rechts	10

S8 Comparison of learning rate with BOpars

level	change	value
2nd level learning rate updates	prevolatile to volatile	0.217
2nd level learning rate updates	volatile to postvolatile	0.059
3rd level learning rate updates	prevolatile to volatile	0.004
3rd level learning rate updates	volatile to postvolatile	0.001

```
## Warning in (function (mapping = NULL, data = NULL, stat = "half_ydensity", :
## Ignoring unknown parameters: `outlier.shape`
```

S9 Posterior predictive checks

```
## Warning: Dropping 'draws_df' class as required metadata was removed.
## Warning: Dropping 'draws_df' class as required metadata was removed.
```

Table S7: Summary of all estimate overlap values

effects	ov
b_diagnosis1	0.9279350
b_diagnosis1:expected1	0.8955151
b_diagnosis1:expected1:phase2	0.8023133
b_diagnosis1:phase2	0.7895629
b_diagnosis1:difficulty1	0.7697581
b_diagnosis1:phase1	0.7656952
b_diagnosis1:expected1:difficulty1	0.7483792
b_phase2	0.7445495
b_phase1	0.6953030
b_diagnosis1:phase2:difficulty2	0.6821911
b_diagnosis1:phase2:difficulty1	0.6646282
b_expected1:phase1:difficulty1	0.6469320
b_diagnosis1:expected1:phase1:difficulty2	0.6183466
b_diagnosis1:expected1:phase1:difficulty1	0.6141940

effects	ov
b_diagnosis1:expected1:phase2:difficulty1	0.6136101
b_expected1:difficulty2	0.6099188
b_expected1:phase1:difficulty2	0.6069053
b_phase1:difficulty1	0.5789345
b_phase2:difficulty2	0.5420668
b_phase2:difficulty1	0.5313039
b_difficulty2	0.5260806
b_expected1:phase2:difficulty1	0.5174780
b_expected1:phase2:difficulty2	0.5034102
b_diagnosis1:difficulty2	0.4636348
b_diagnosis1:expected1:phase2:difficulty2	0.4329468
b_diagnosis1:expected1:difficulty2	0.4211235
b_diagnosis1:phase1:difficulty2	0.4065543
b_diagnosis1:expected1:phase1	0.4047931
b_expected1:difficulty1	0.3869115
b_diagnosis1:phase1:difficulty1	0.3710294
b_phase1:difficulty2	0.3059093
b_expected1:phase1	0.2467234
b_expected1	0.1858033
b_expected1:phase2	0.0313613
b_difficulty1	0.0076268

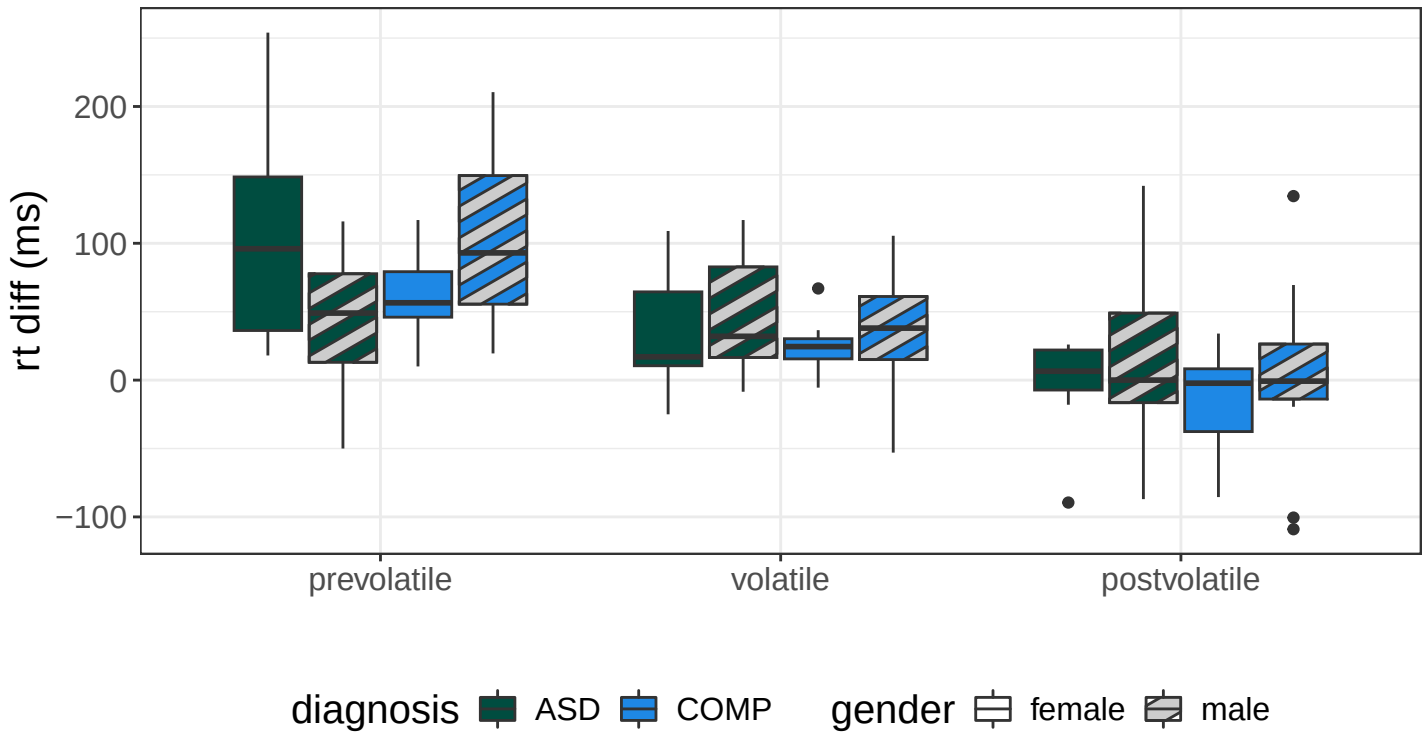
Table S8: Summary of estimate overlap values pertaining to groups

mean	sd	ci.low	ci.high
54.46	21.87	2.78	90.04

mean	sd	ci.low	ci.high
63.29	17.96	38.54	91.42

S10 Exploration of gender differences

S10.1 Reaction times: expectancy effect



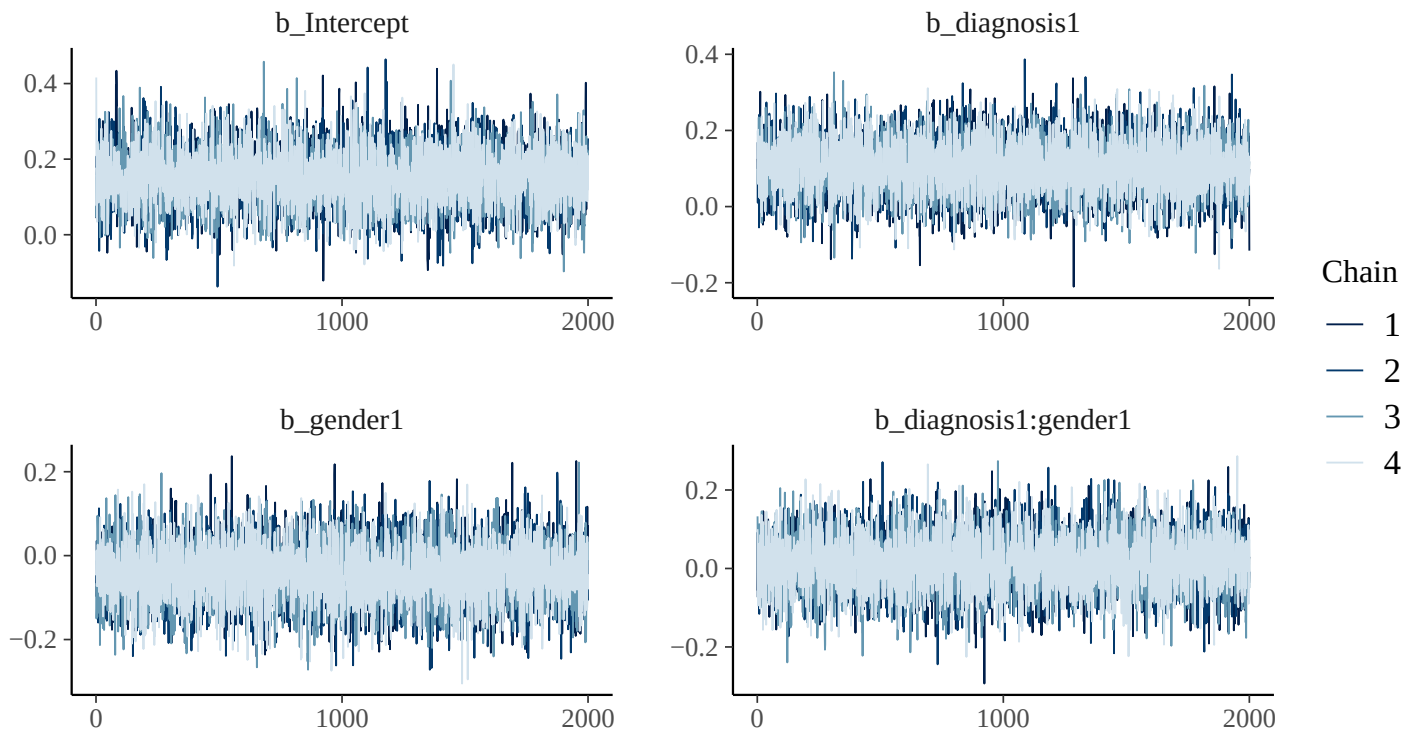
S10.2 HGF parameters

```
##      [,1]
## ASD      1
## COMP     -1

##      [,1]
## female    1
## male      -1
```

S10.2.1 Phasic environmental volatility

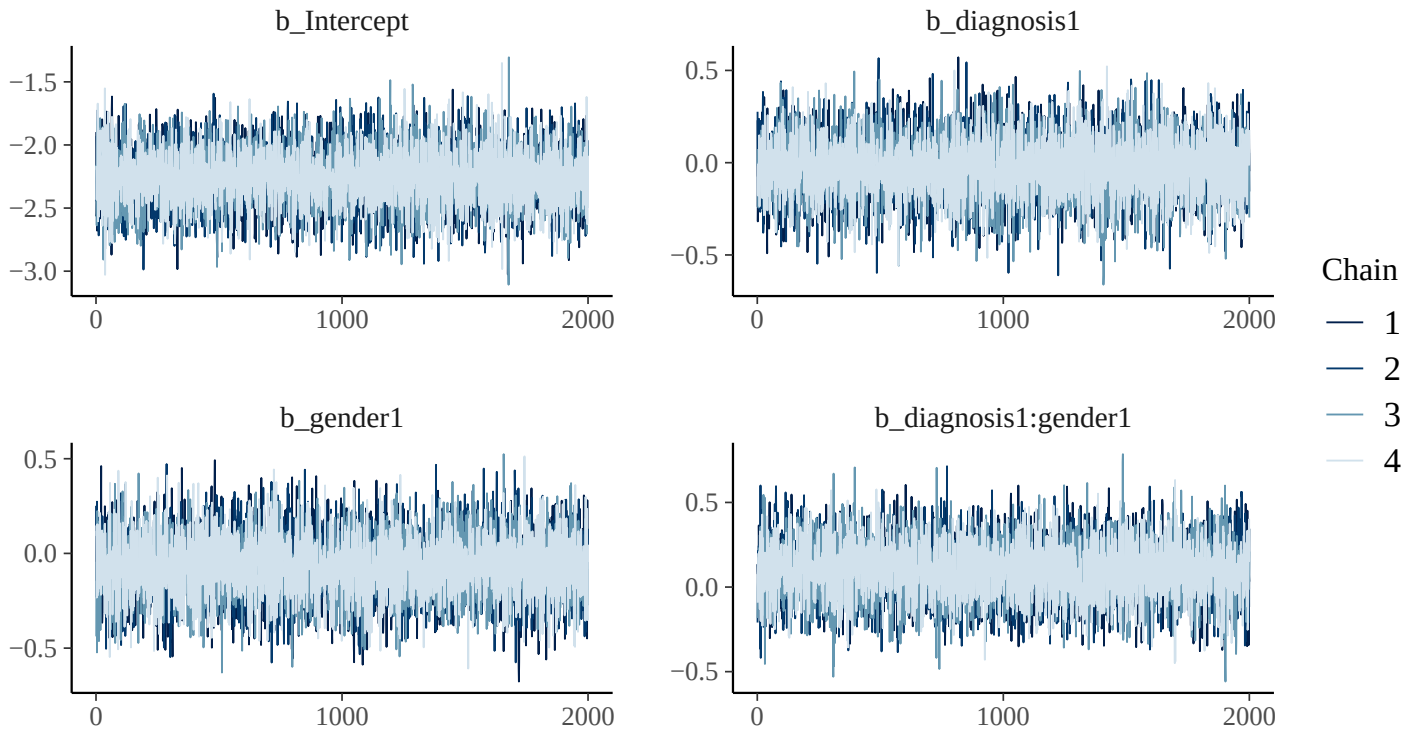
```
##
## Divergences:
##
## Tree depth:
##
## Energy:
## [1] 0
```



```
## Family: gaussian
## Links: mu = identity
## Formula: be4 ~ diagnosis * gender
## Data: df.hgf (Number of observations: 44)
## Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
##         total post-warmup draws = 8000
##
## Regression Coefficients:
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept          0.15     0.07    0.01    0.30 1.00    10401    6015
## diagnosis1          0.10     0.07   -0.03    0.24 1.00    10182    6268
## gender1            -0.04     0.07   -0.18    0.10 1.00    10729    6306
## diagnosis1:gender1  0.01     0.07   -0.12    0.15 1.00     8429    5571
##
## Further Distributional Parameters:
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma          0.49     0.06    0.39    0.61 1.00     8185    5672
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

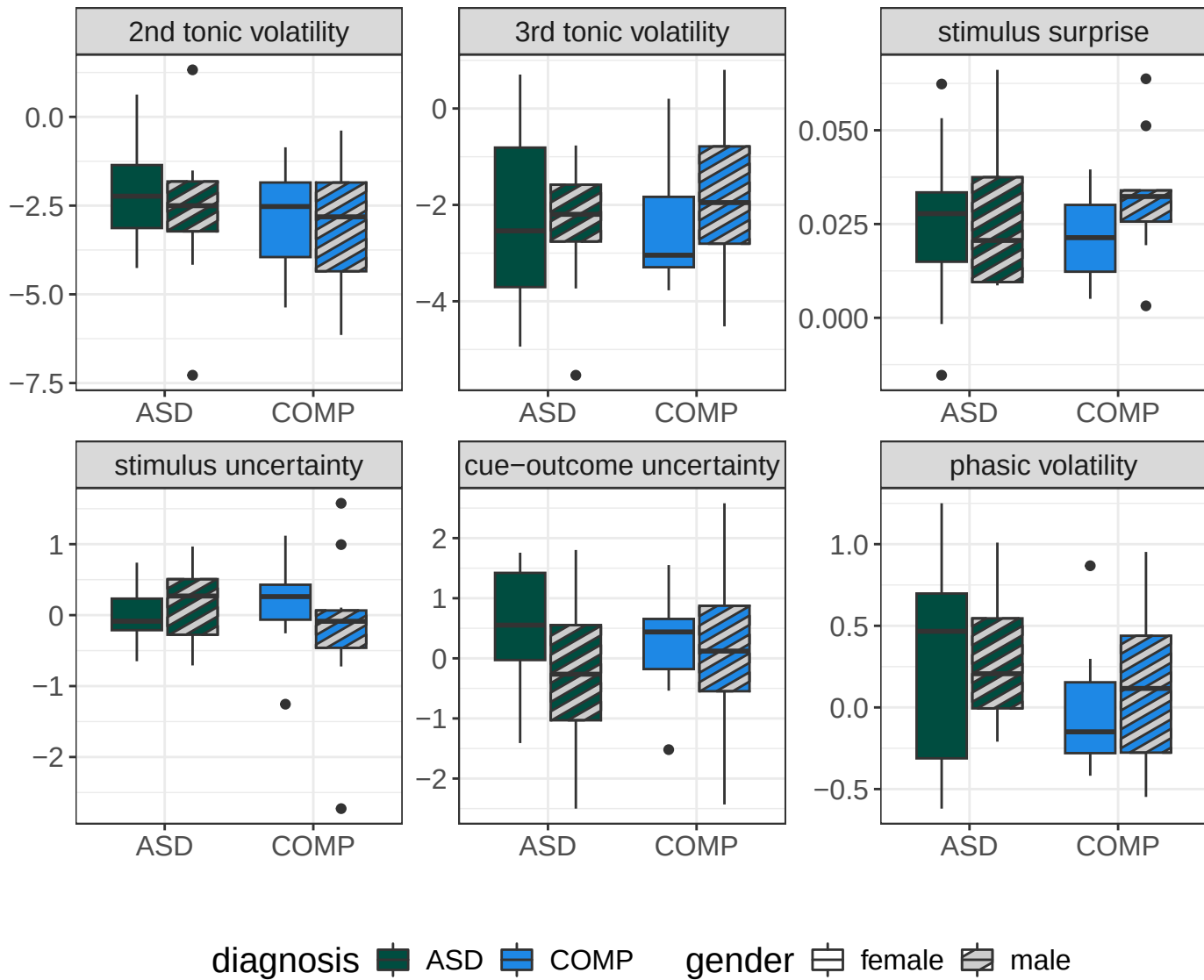
S10.2.2 Tonic environmental volatility

```
##
## Divergences:
##
## Tree depth:
##
## Energy:
## [1] 0
```



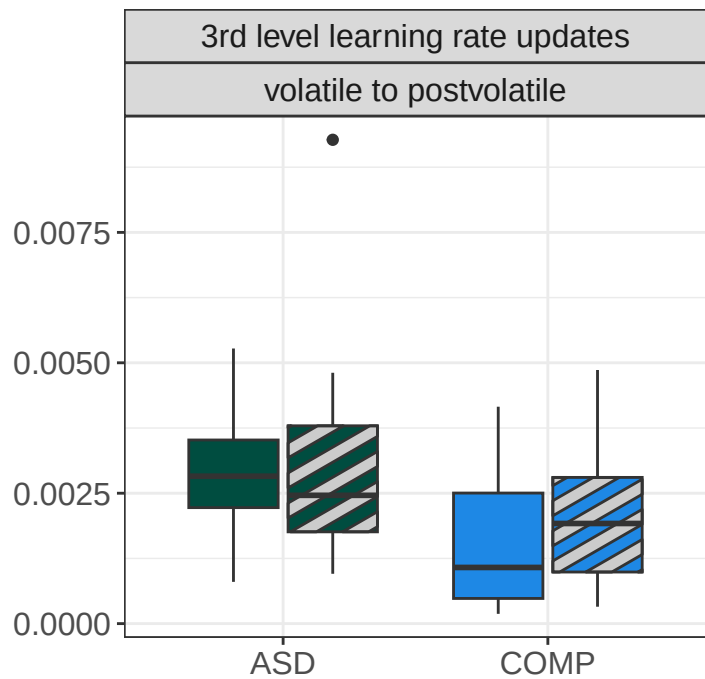
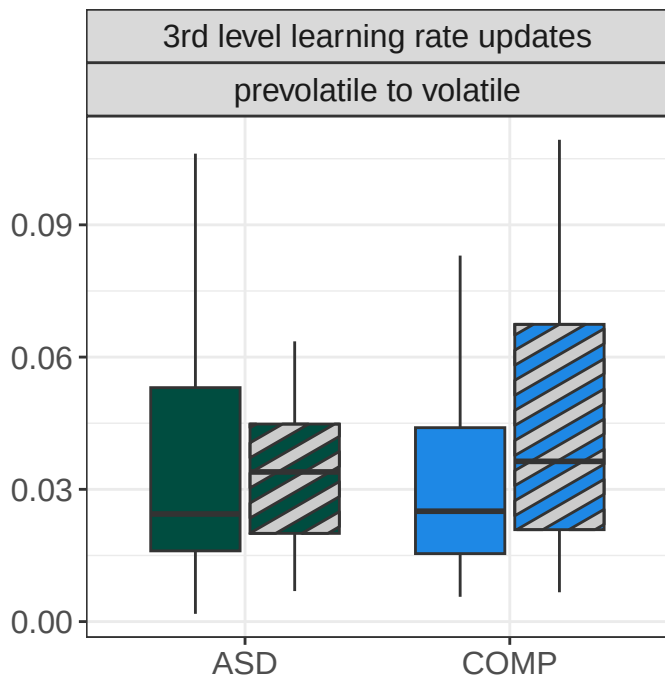
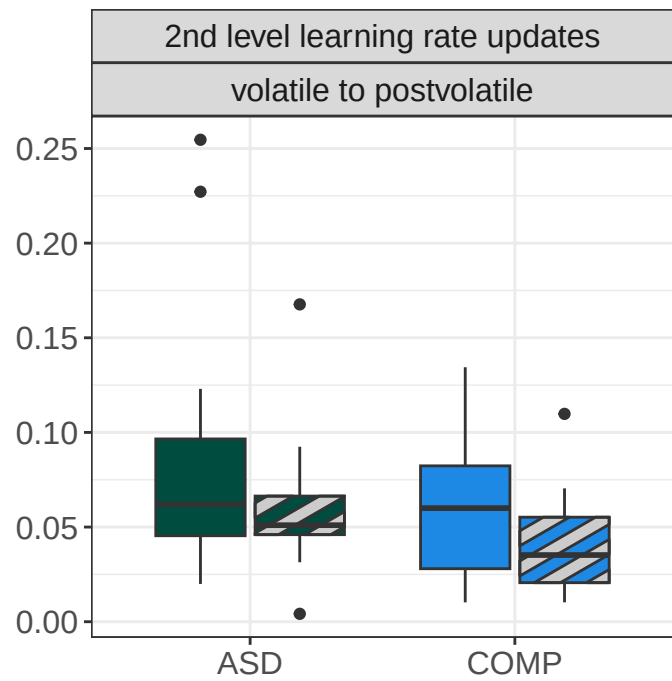
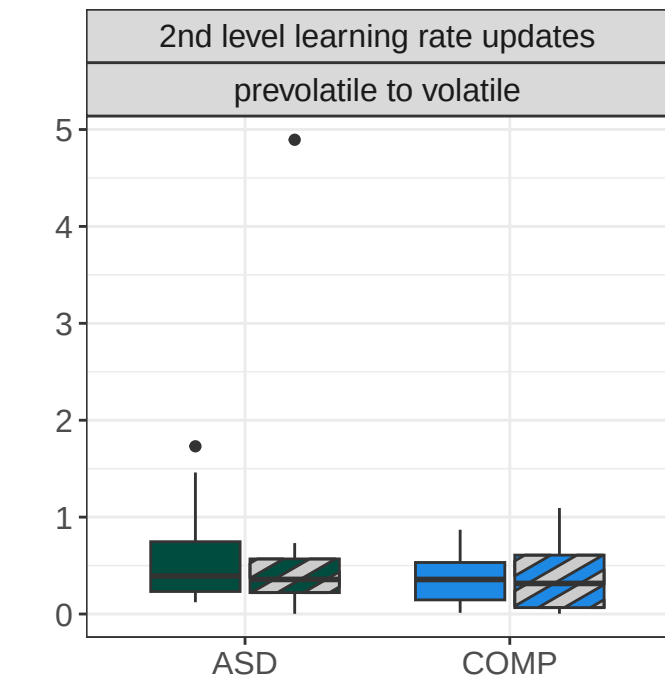
```
## Family: gaussian
## Links: mu = identity
## Formula: om3 ~ diagnosis * gender
## Data: df.hgf (Number of observations: 44)
## Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
##         total post-warmup draws = 8000
##
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept          -2.26     0.22   -2.70   -1.82 1.00     9976     6020
## diagnosis1          -0.03     0.16   -0.35    0.29 1.00     9658     5759
## gender1             -0.07     0.16   -0.39    0.26 1.00     9829     6193
## diagnosis1:gender1    0.10     0.17   -0.23    0.42 1.00    10245     6105
##
## Further Distributional Parameters:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma         1.46     0.14     1.22     1.75 1.00     9158     6150
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

S10.2.3 Plot of all HGF parameters



S10.3 Learning rate updates

	bf	bf.diff	bf.int
level + change	105.331	0.010	anecdotal evidence in favour of
level + change + level:change	105.321	0.630	anecdotal evidence in favour of
diagnosis + level + change + level:change	104.691	0.011	anecdotal evidence in favour of
diagnosis + level + change	104.679	0.862	anecdotal evidence in favour of
gender + level + change + level:change	103.818	0.040	anecdotal evidence in favour of
gender + level + change	103.777	0.124	anecdotal evidence in favour of



diagnosis  ASD  COMP gender  female  male